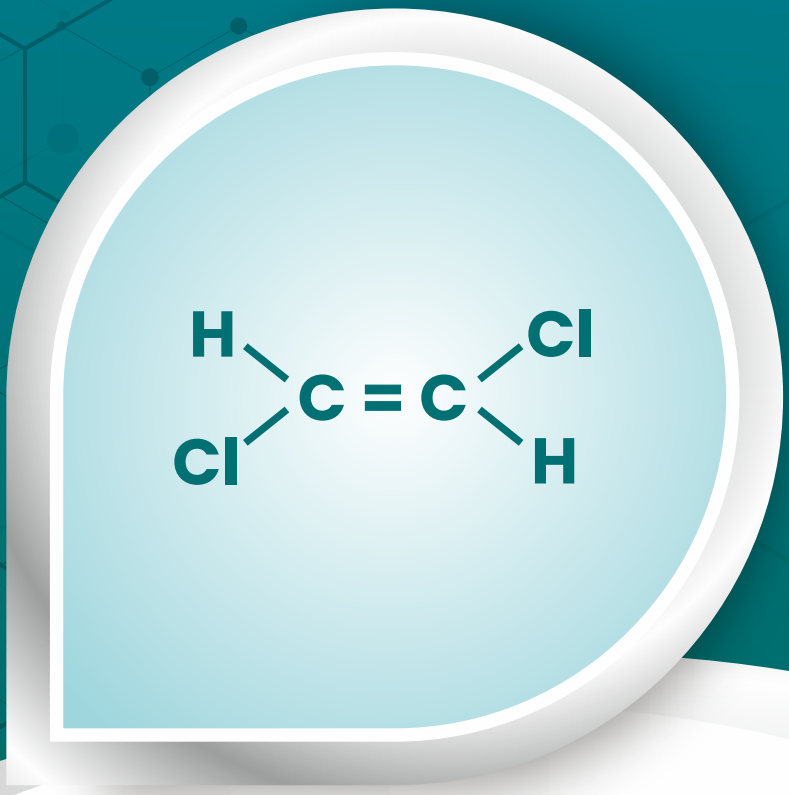
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)

path to success

ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE



Exercise

Isomerism

(English Medium)

ISOMERISM

EXERCISE # O-I (MAINS ORIENTED)

Single Correct Type :

1. Minimum number of carbon atom required by a Hydrocarbon alkane, alkene and alkyne to show chain isomerism respectively are :

(A) 4, 4, 4 (B) 4, 3, 3 (C) 4, 4, 3 (D) 4, 4, 5

SR0001

2. C_7H_7Cl shows how many benzenoid aromatic isomers ?

(A) 4 (B) 3 (C) 5 (D) 6

SR0002

3. How many structural isomers of primary amines are possible for the formula $C_4H_{11}N$?

(A) 2 (B) 3 (C) 4 (D) 5

SR0003

4. How many benzenoid structure are possible for molecular form C_7H_8O :

(A) 3 (B) 4 (C) 5 (D) 6

SR0004

5. Which of the following compound give geometrical isomerism ?

(A) CH_2Cl_2 (B) $CH_2Cl—CH_2Cl$ (C) $CHBr = CHCl$ (D) $CH_2Cl—CH_2Br$.

SR0005

6. Maximum potential energy is associated with which of the following conformers of n-butane :

(A) Anti (B) Gauche (C) Eclipsed (D) Partial eclipsed

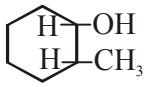
SR0006

7. Which among the following compounds will show geometrical isomerism :

(A) $CH_3–CH=CH_2$ (B) $CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CH_2$ (C) $CH_3–\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=CHD$ (D) $CH_3–CH=CHD$

SR0007

8. Which among the following will show geometrical isomerism :

(A)  (B)  (C) $C_6H_5–CH=N–OH$ (D) All of these

SR0008

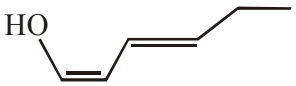
9. Which one of the following is Z isomer :

(A) $\begin{matrix}CH_3\\H\end{matrix}\!\!>\!C=C\!<\!\begin{matrix}Br\\Cl\end{matrix}$ (B) $\begin{matrix}CH_3\\H\end{matrix}\!\!>\!C=C\!<\!\begin{matrix}Cl\\Br\end{matrix}$ (C) $\begin{matrix}CH_3\\Br\end{matrix}\!\!>\!C=C\!<\!\begin{matrix}Cl\\H\end{matrix}$ (D) $\begin{matrix}CH_3\\H\end{matrix}\!\!>\!C=C\!<\!\begin{matrix}H\\CH_3\end{matrix}$

SR0009

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

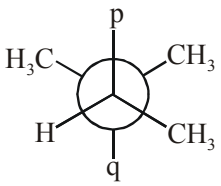
10. The correct stereochemical descriptions for the structure given below are :



(A) 1 Z, 3Z (B) 1Z, 3E (C) 1E, 3E (D) 1E, 3Z

SR0010

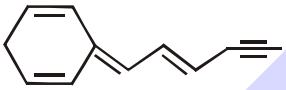
11. What is the value of p & q of following conformer of 2,3-dimethyl butane ?



(A) p = H, q = H (B) p = CH_3, q = CH_3 (C) p = CH_3, q = H (D) p = H, q = CH_3

SR0011

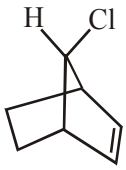
12. The number of cis-trans isomer possible for the following compound.

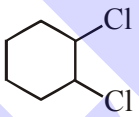


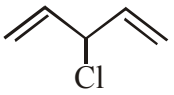
(A) 2 (B) 4 (C) 6 (D) 8

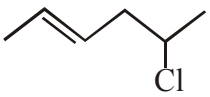
SR0012

13. Which of the following can not show geometrical isomerism :

(A) 

(B) 

(C) 

(D) 

SR0013

14. How many structural formula are possible when one of the hydrogen is replaced by a chlorine atom in benzene & naphthalene respectively ?

(A) 1, 1 (B) 1,2 (C) 2, 2 (D) 2, 1

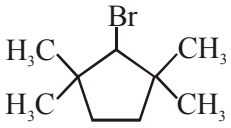
SR0076

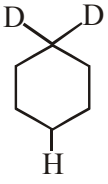
15. The number of structural isomers of C_5H_{10} is :-

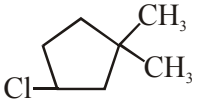
(A) 10 (B) 11 (C) 12 (D) 13

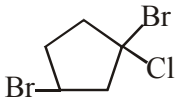
SR0015

16. Which of the following compounds will show geometrical isomerism.

(A) 

(B) 

(C) 

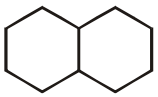
(D) 

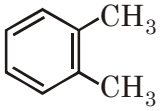
SR0016

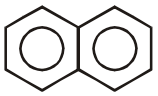
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

17. Which of the following will show geometrical isomerism?

(A)  Benzene

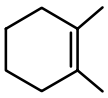
(B)  Decalin

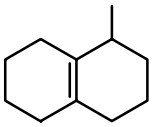
(C)  o–Xylene

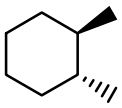
(D)  Naphthalene

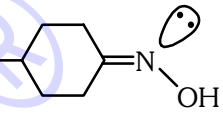
SR0077

18. Which of the following cannot be represent in E or Z

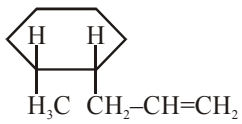
(A) 

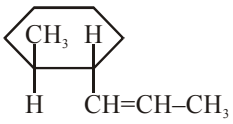
(B) 

(C) 

(D) 

SR0018

19. 



(A) Geometrical isomer

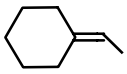
(B) Ring chain isomer

(C) Position isomer

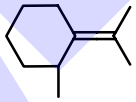
(D) Chain isomer

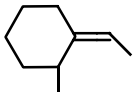
SR0019

20. Which of the following can show geometrical isomesrism :

(A) 

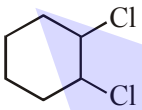
(B) 

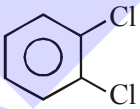
(C) 

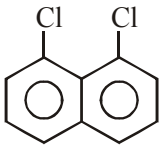
(D) 

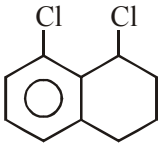
SR0020

21. Which of the following will show Geometrical isomerism.

(A) 

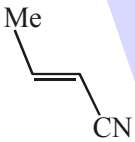
(B) 

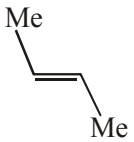
(C) 

(D) 

SR0021

22. Which of the following will have lowest dipole moment?

(A) 

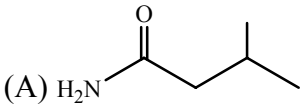
(B) 

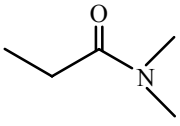
(C) 

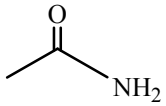
(D) 

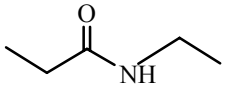
SR0022

23. Which of the following compounds can show geometrical isomerism in its resonating structure.

(A) 

(B) 

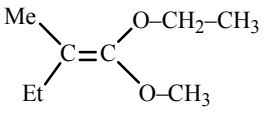
(C) 

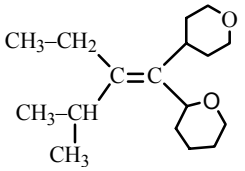
(D) 

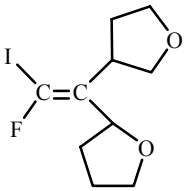
SR0023

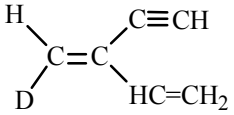
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

24. Which of the following represent Z-isomer ?

(A) 

(B) 

(C) 

(D) 

SR0024

25. Which of the following statement is not correct :-

(A) Cyclobutane is a planar compound

(B) Trans cyclohexadecene is relatively more stable than its cis form

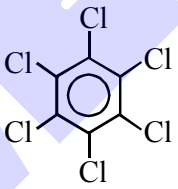
(C) Cis form of 1,3,5-trimethylclohexane is relatively more stable than its trans form

(D) Cis 1,2-dichloroethene is relatively more stable than its trans form.

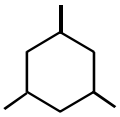
SR0025

26. Select which one of the following can not show geometrical isomerism

(A) 

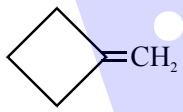
(B) 

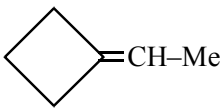
(C) Me–CH = C = C = CH – Me

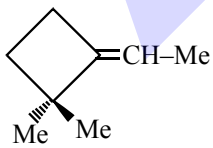
(D) 

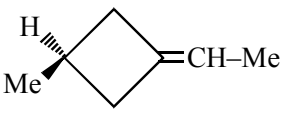
SR0026

27. Which one of the following is capable of showing geometrical isomerism.

(A) 

(B) 

(C) 

(D) 

SR0027

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\65

More than one correct Type :

28. Which of the following statements is/are not correct :

(A) Metamerism belongs to the category of structural isomerism

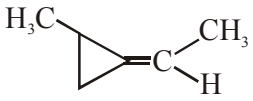
(B) Tautomeric structures are the resonating structures of a molecule

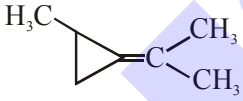
(C) Keto form is always more stable than the enol form

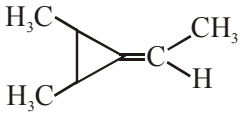
(D) Geometrical isomerism is shown only by alkenes

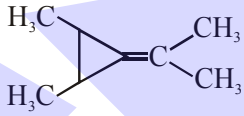
SR0028

29. Which of the following will show geometrical isomerism :

(A) 

(B) 

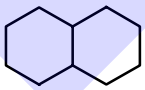
(C) 

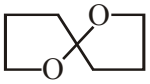
(D) 

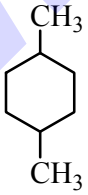
SR0029

30. Which of the following compound will shows geometrical isomerism ?

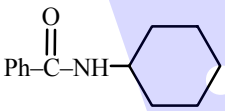
(A) Ph –CH =N –Me

(B) 

(C) 

(D) 

SR0030

31 

(A) This molecule shows geometrical isomerism.

(B) One of the resonating structure of this molecule shows geometrical isomerism

(C) One of the tautomer of this molecule shows geometrical isomerism

(D) In acidic medium this molecule shows geometrical isomerism.

SR0031

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

Assertion / Reasoning Type :

32. **Statement-1 :** All double bond containing compounds show geometrical isomerism.

and

Statement-2 : Alkenes have restricted rotation about the carbon-carbon double bond.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

SR0032

33. **Statement-1 :** According to CIP sequence rule the priority of the groups is

$-CH = CH_2 < -C \equiv CH < -C \equiv N < -CH = O$

and

Statement-2 : Priority of the given groups are based on molecular mass of groups.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

SR0033

Comprehension Type :

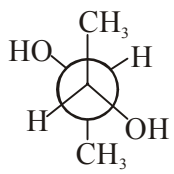
Paragraph for Q. 34 to 35

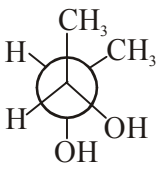
Different spatial arrangements of the atoms that result from rotation about a single bond are conformers. n-Butane has four conformers eclipsed, partial eclipsed, gauche and anti. The stability order of these conformers are as follows :

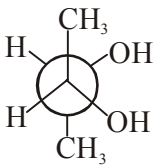
Anti > gauche > Partial eclipsed > Fully eclipsed

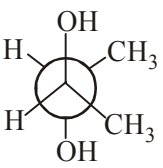
Although anti is more stable than gauche but in some cases gauche is more stable than anti.

34. Which one of the following is the most stable conformer ?

(A) 

(B) 

(C) 

(D) 

SR0034

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

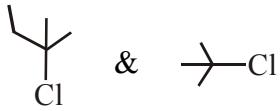
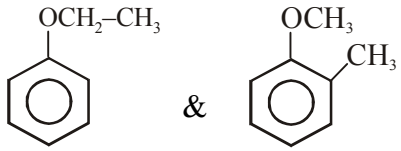
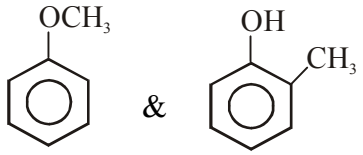
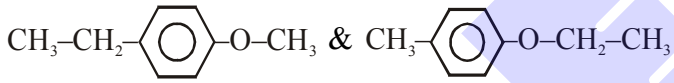
35. Number of possible conformations of n-butane is :

(A) 2 (B) 4 (C) 6 (D) infinite

SR0034

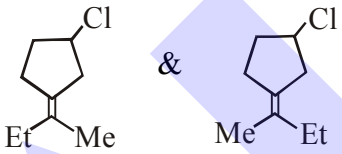
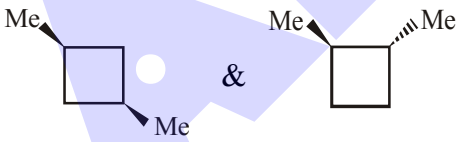
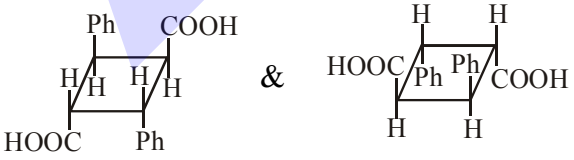
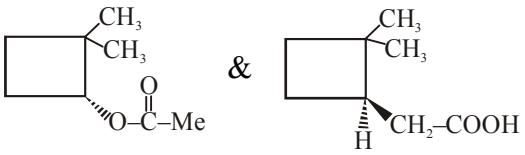
Matrix Match Type :

36. Match the column I with column II.

Column-I (reaction)	**Column-II (stereoisomers)**
(A) 	(P) Homologs
(B) 	(Q) Functional isomers
(C) 	(R) Metamer
(D) 	(S) Position isomers

SR0035

37. Match the column I with column II.

Column-I	**Column-II**
(A) 	(P) Identical
(B) 	(Q) Functional isomers
(C) 	(R) Geometrical Isomers
(D) 	(S) Positional Isomers

SR0036

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

Subjective Type :

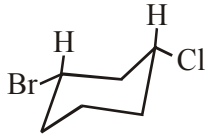
38. Considering rotation about the C_3–C_4 bond of 2-methylhexane :

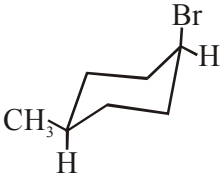
(a) Draw the Newman projection of the most stable conformer

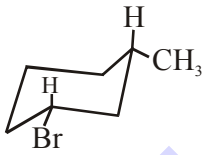
(b) Draw the Newman projection of the least stable conformer

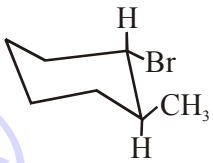
SR0037

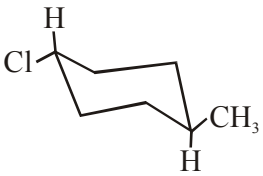
39. Determine whether each of the following compounds is a cis isomer or a trans isomer.

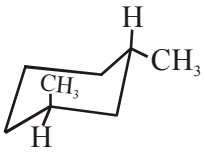
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

SR0038

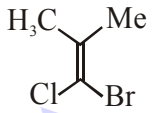
40. Draw the most stable conformer of N-methylpiperidine.

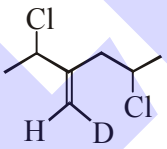
SR0039

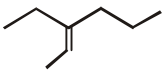
41. How many pair(s) of geometrical isomers are possible with C_6H_{12} (only in open chain structures)

SR0040

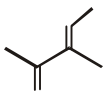
42. Identify molecules which usually can show geometrical isomerism (at room temperature):

(I) 

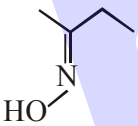
(II) 

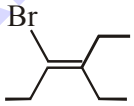
(III) 

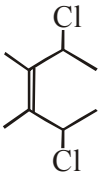
(IV) 

(V) 

SR0041

(VI) 

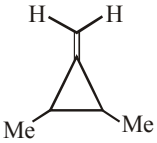
(VII) 

(VIII) 

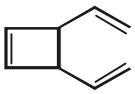
(IX) 

SR0042

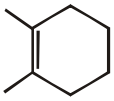
(X) 

(XI) 

(XII) 

(XIII) 

(XIV) 

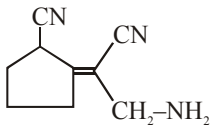
(XV) 

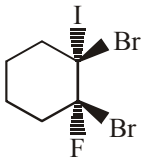
SR0043

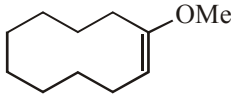
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\65

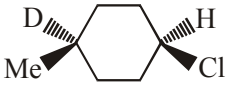
43. Calculate the number of Benzenoid isomers possible for C_6H_3ClBrI.

SR0044

44. (a) 

(b) 

(c) 

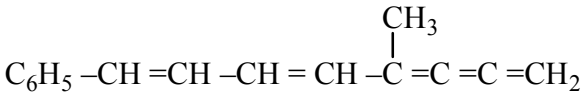
(d) 

Assign E or Z to the following compounds and write 1 for E and 2 for Z.

Write answer of part (a), (b), (c) & (d) in the same order and present the four digit number as answer in OMR sheet. For example: If all these answer are 9 then fill 9999 in OMR sheet.

SR0045

45. (a) Number of geometrical isomers of the following compound.



(b) Total number 2° and 3° Alcohols possible for $C_5H_{11}OH$. without connting stereo isomers.

Write answer of part (a) & (b) in the same order and present the four digit number as answer in OMR sheet. For example: If your Answer for (a) is 9 & (b) is 9 then fill 0909 in OMR sheet.

SR0046

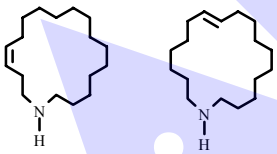
46. Analyse the following pairs of compounds.

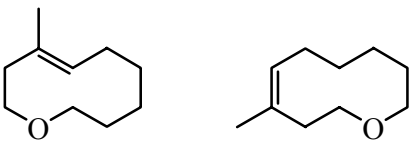
Write 1 if they are Geometrical isomers.

Write 2 if they are Chain isomers.

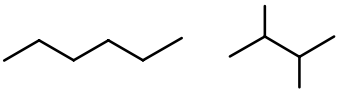
Write 3 if they are position isomers.

Write 4 if they are Functional group isomers.

(a) 

(b) 

(c) 

(D) 

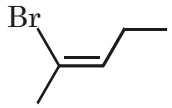
Write answer of part (a), (b), (c) & (d) in the same order and present the four digit number as answer in OMR sheet. For example: If all these answers are 9 then fill 9999 in OMR sheet.

SR0047

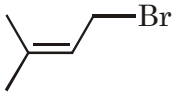
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

STEREOISOMERISM

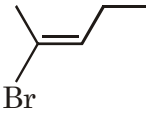
47. Which one of the following statements concerning compounds **V**–**Z** is true :



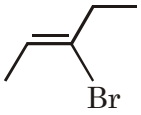
(V)



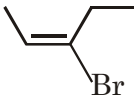
(W)



(X)



(Y)



(Z)

(A) **V** and **X** are conformational isomers
(B) **Y** and **Z** are constitutional isomers
(C) **X** and **Y** are constitutional isomers
(D) **V** and **Y** are stereoisomers

SE0001

48. Which of the following compound has no isomer?

(A) $CH_3CH_2CH_2Cl$ (B) CH_3CHO
(C) CH_2=CH–Cl (D) $ClCH_2CH_2Cl$

SE0002

49. Compound CH_2Cl_2 contain :

(A) Plane of symmetry (B) Centre of symmetry
(C) Axis of symmetry (D) Both (A) & (C)

SE0003

50. Number of POS present in CH_4 :

(A) 3 (B) 4 (C) 5 (D) 6

SE0004

51. How many stereoisomers of the following molecule are possible ?

HOOC.CH=C=CH.COOH

(A) Two optical isomers (B) Two geometrical isomers
(C) Two optical and two geometrical isomers (D) None

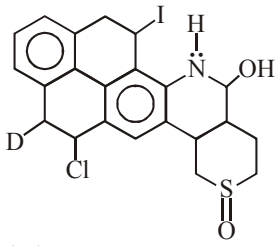
SE0005

52. Which of the following compounds exhibits stereoisomerism–

(A) 2–Methylbutene–1 (B) 3–Methylbutyne–1
(C) 3–Methylbutanoic acid (D) 2–Methylbutanoic acid

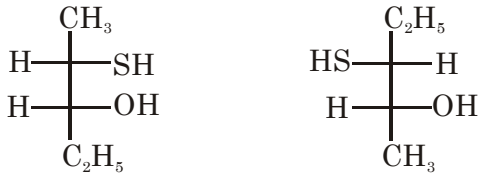
SE0129

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp!, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

53.  has 'x' chiral centre then find the value of x :

(A) 7 (B) 8 (C) 6 (D) 5

SE0007

54. 

(A) Diastereomers (B) Enantiomers (C) Identical (D) Constitutional isomers

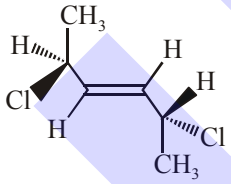
SE0008

55. The number of optically active compounds in the isomers of C_4H_9Br is :

(A) 1 (B) 2 (C) 3 (D) 4

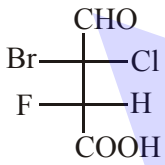
SE0009

56. Compound have :



(A) Plane of symmetry (B) Centre of symmetry
(C) Axis of symmetry (D) None

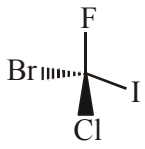
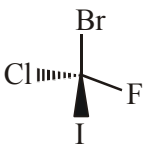
SE0010

57. 

Configuration of compound is :

(A) 2S, 3S (B) 2R , 3S (C) 2R , 3R (D) 2S , 3R

SE0011

58.  &  have configuration respectively :

(A) R, S (B) S, S (C) S, R (D) R, R

SE0012

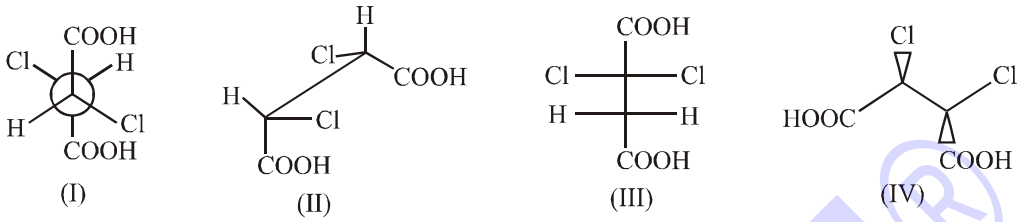
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

59. Minimum molecular weight of a hydrocarbon containing minimum number of C-atom to show optical isomerism :

(A) 100 (B) 80 (C) 68 (D) 70

SE0013

60. For the given configuration :

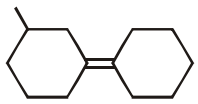


Which of the compound/configuration are optically active :

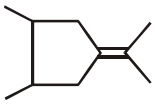
(A) I (B) II (C) III (D) IV

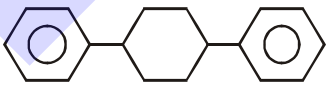
SE0014

61. Compounds which can show both optical as well as geometrical isomerism :

(A) 

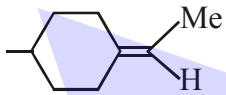
(B) 

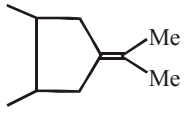
(C) 

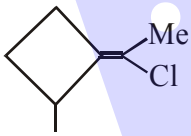
(D) 

SE0015

62. Which of the following will not show optical isomerism :

(A) 

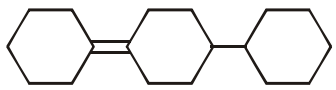
(B) 

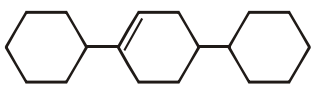
(C) 

(D) 

SE0016

63. Optical & geometrical isomerism both can be shown by :-

(A) 

(B) 

(C) 

(D) 

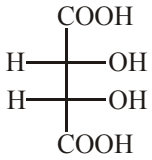
SE0017

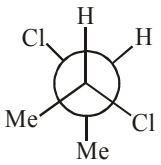
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

64. Which of the following will not show optical isomerism :

(A) Cl–CH=C=C=CH–Cl

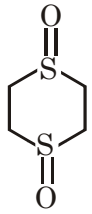
(B) Cl–CH=C=C=C=CH–Cl

(C) 

(D) 

SE0018

65. The correct statement for the given compound is



(A) It can shows geometrical isomerism

(B) It can show optical isomerism

(C) It contain chiral centre

(D) None of these

SE0019

66. Meso-tartaric acid and d-tartaric acid are :-

(A) Positional isomers

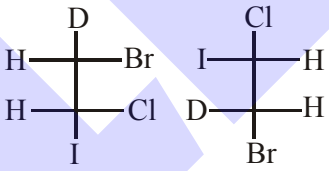
(B) Enantiomers

(C) Diastereomers

(D) Racemic mixture

SE0020

67. The two compounds given below are :



(A) Enantiomers

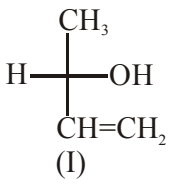
(B) Diastereomers

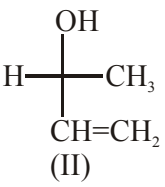
(C) Optically inactive

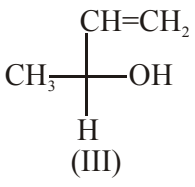
(D) Identical

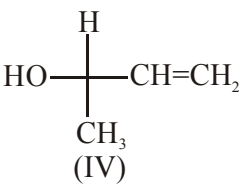
SE0021

68. Which of the following combinations amongst the four Fischer projections represents the same absolute configurations ?


(I)


(II)


(III)


(IV)

(A) (II) and (III)

(B) (I) and (IV)

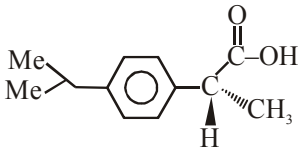
(C) (II) and (IV)

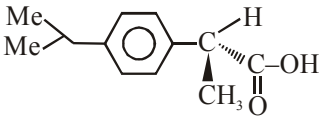
(D) (III) and (IV)

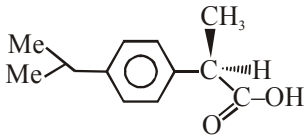
SE0022

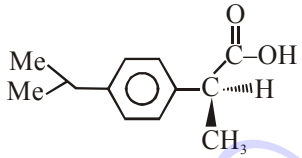
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

69. The S-ibuprofen is responsible for its pain relveing property. Which one of the structure shown is S-ibuprofen :

(A) 

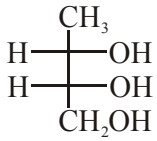
(B) 

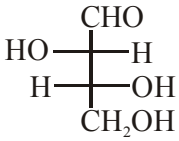
(C) 

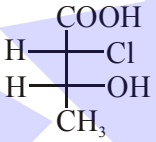
(D) 

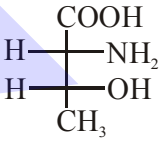
SE0023

70. Which of the following is a 'threo' isomer :

(A) 

(B) 

(C) 

(D) 

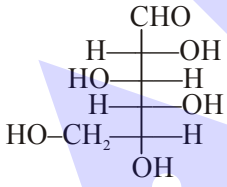
SE0024

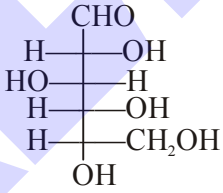
71. Number of possible stereoisomers of glucose are :-

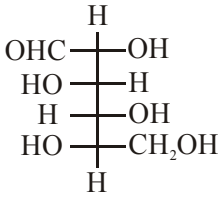
(A) 10 (B) 8 (C) 16 (D) 20

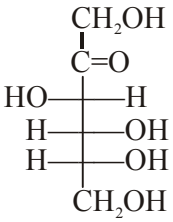
SE0025

72. Which of the following is not D sugar :

(A) 

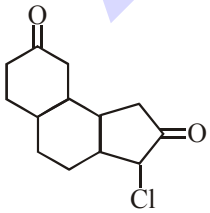
(B) 

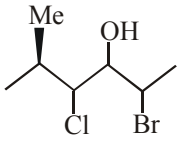
(C) 

(D) 

SE0026

73. Number of chiral centres in [X] & [Y] is a & b respectively. The value of (a–b) is :

[X] = 

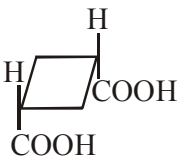
[Y] = 

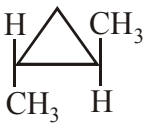
(A) 1 (B) 2 (C) 3 (D) 4

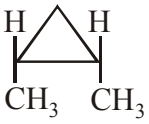
SE0027

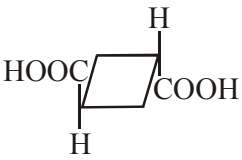
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

74. Which one of the following is resolvable :

(A) 

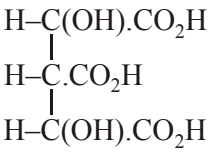
(B) 

(C) 

(D) 

SE0028

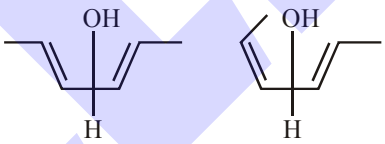
75. How many stereoisomers can exist for the following acid.



(A) Two (B) Four (C) Eight (D) Six

SE0029

76. Incorrect relationship between given compounds are



(A) Both are geometrical isomers

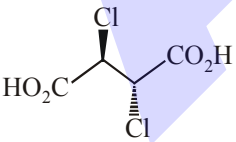
(B) Both are stereo isomers

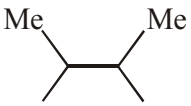
(C) Both are enantiomers

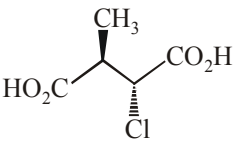
(D) Both are diastereomers

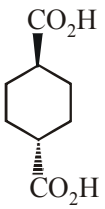
SE0030

77. Identify meso compound.

(A) 

(B) 

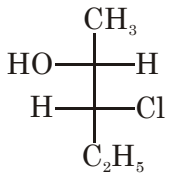
(C) 

(D) 

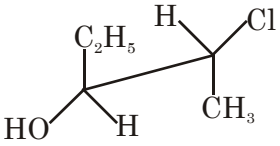
SE0031

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

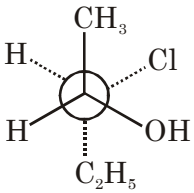
78. The two projection formulae that represent a pair of enantiomers are :-



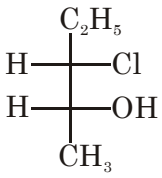
(I)



(II)



(III)



(IV)

(A) I and II (B) III and IV

(C) I and III (D) II and IV

SE0032

79. A pure sample of 2-chlorobutane shows rotation of PPL by 30° in standard conditions. When above sample is made impure by mixing its opposite form, so that the composition of the mixture becomes 87.5% d-form and 12.5% ℓ-form, then what will be the observed rotation for mixture.

(A) – 22.5° (B) + 22.5° (C) + 7.5° (D) – 7.5°

SE0033

80. When an optically active compound is placed in a 10 dm tube is present 20 gm in a 200 ml solution rotates the PPL by 30°. Calculate the angle of rotation & specific angle of rotation if above solution is diluted to 1 Litre.

(A) 16° & 36° (B) 6° & 30° (C) 3° & 30° (D) 6° & 36°

SE0034

81. Identify % optical purity if 6 gm (+)-2-butanol is mixed with 2 gm (–)-2-butanol.

(A) 50 % (B) 66.6 % (C) 33.3 % (D) 75 %

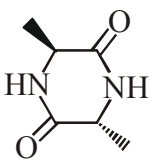
SE0035

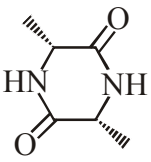
82. A mixture of d and ℓ, 2-bromobutane contain 75% d-2-bromobutane. Calculate enantiomeric excess.

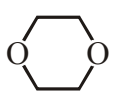
(A) 75% (B) 25% (C) 50% (D) 100%

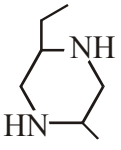
SE0036

83. Which of the following is example of meso compound?

(A) 

(B) 

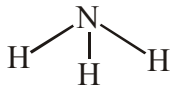
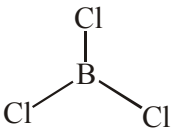
(C) 

(D) 

SE0037

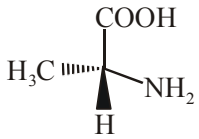
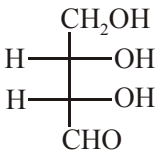
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

84. Which of the following has C_2 & C_3 axis of symmetry ?

(A)  (B) H_2CCl_2 (C) $H_3C–Cl$ (D) 

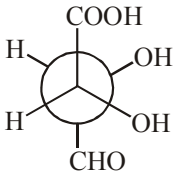
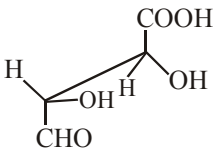
SE0038

85. Configuration of I & II respectively will be :

(A) D, D (B) L , D (C) D , L (D) L, L

SE0039

86.   , compound related as :

(A) Enantiomers (B) Conformation (C) Identicle (D) Diastereomers

SE0040

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\ModuleNome, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

EXERCISE # O-II (STEREO ISOMERISM)

Single correct Option Type :

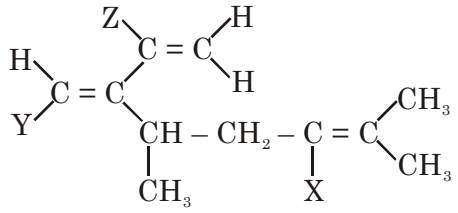
1. Molecular formula $C_5H_{10}O$ can have :

(A) 6-Aldehyde, 4-Ketone (B) 5-Aldehyde, 3-Ketone

(C) 4-Aldehyde, 3-Ketone (D) 5-Aldehyde, 2-Ketone

SE0041

2. In the given halogenoalkene M, atoms X, Y and Z represents hydrogen or bromine or chlorine. To show cis-trans isomerism, what could be the identities of atoms X, Y and Z?

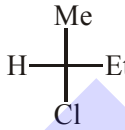


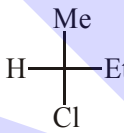
(M)

	X	Y	Z
1	Cl	H	Br
2	H	Br	Cl
3	Cl	Br	H

(A) 1, 2 and 3 (B) 1 and 2 only (C) 2 and 3 only (D) 1 and 3 only

SE0042

3. **Statement–1 :**  is a chiral resolvable molecule.

Statement–2 :  is non-superimposable on its mirror image.

(A) Statement–1 is true, Statement–2 is true; Statement–2 is not the correct explanation of Statement–1

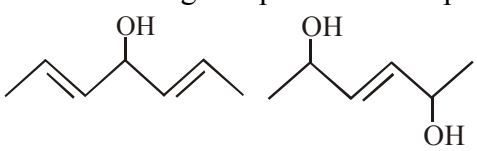
(B) Statement–1 is true, Statement–2 is true ; Statement–2 is the correct explanation of Statement–1

(C) Statement–1 is true, Statement–2 is false

(D) Statement–1 is false, Statement–2 is true

SE0043

4. Total number of stereoisomer of following compounds are respectively :-



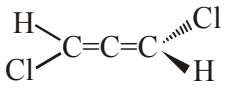
(A) 4, 6 (B) 8 (C) 6,6 (D) 8, 8

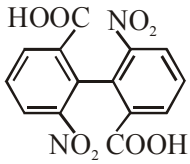
SE0044

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\65

5. Which of the following compounds are optically active ?

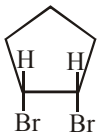
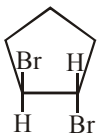
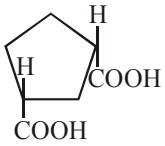
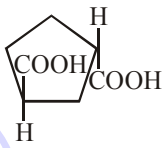
(A) $CH_3.CHOH.CH_2.CH_3$ (B) $H_2C=CH.CH_2.CH=CH_2$

(C) 

(D) 

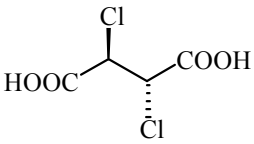
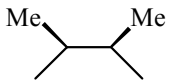
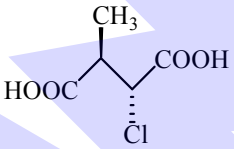
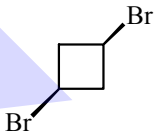
SE0045

6. Which out the following are Non-resolvable :

(A)  (B)  (C)  (D) 

SE0046

7. Identify compound(s) which is/are not meso :

(A)  (B)  (C)  (D) 

SE0047

8. Which of the following statements for a meso compound is/are correct :

(A) The meso compound has either a plane or centre of symmetry

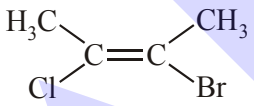
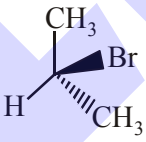
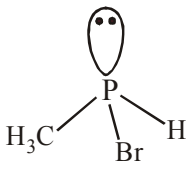
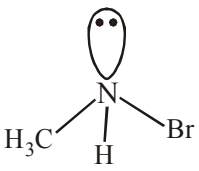
(B) The meso compound is optically inactive due to internal compensation.

(C) The meso compound is achiral

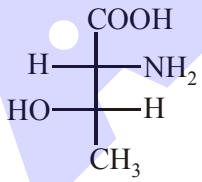
(D) The meso compound is formed when equal amounts of two enantiomers are mixed

SE0048

9. Among the following the non- resolvable compound is/are :

(A)  (B)  (C)  (D) 

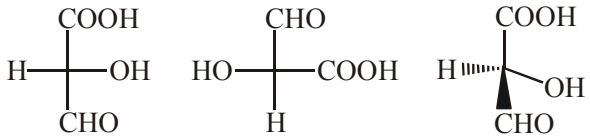
SE0049

10. Compound  is/are ?

(A) (2R, 3S), L (B) L, Erythreo (C) Threo , D (D) (2R, 3S), D

SE0050

11. Relation between compounds are :

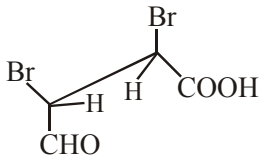


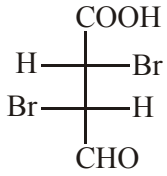
(A) I & II = Enantiomers (B) II & III = Enantiomers

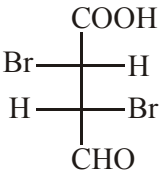
(C) I & II = Identical (D) II & III = Identical

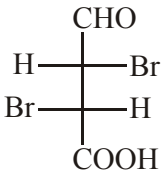
SE0051

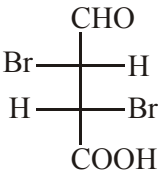
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

12.  has fisher diagram ?

(A) 

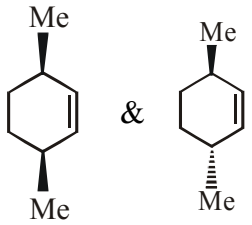
(B) 

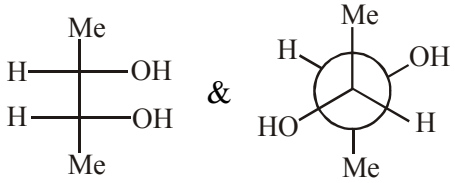
(C) 

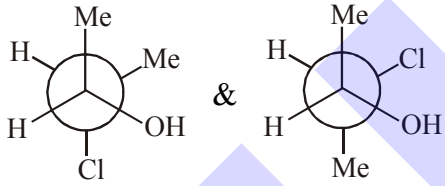
(D) 

SE0052

13. Identify correct relation between pair of compounds ?

(A)  Diastereomers

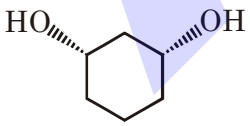
(B)  Identical

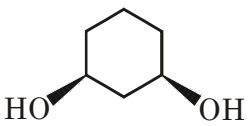
(C)  Diastereomers

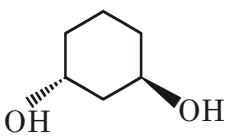
(D)  Enantiomers

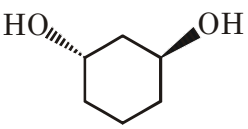
SE0053

14. Which two of the following compounds represents a pair of enantiomers?

(I) 

(II) 

(III) 

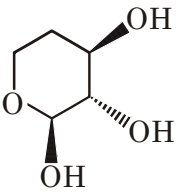
(IV) 

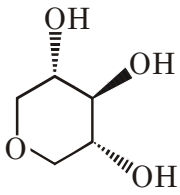
(A) I & II (B) II & III (C) III & IV (D) II & IV

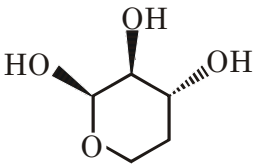
SE0054

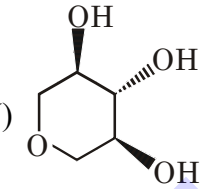
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

15. Which two of the following compounds are diastereomers?

(I) 

(II) 

(III) 

(IV) 

(A) I & II (B) II & IV (C) III & IV (D) I & III

SE0055

16. The correct relation between the following compounds is :

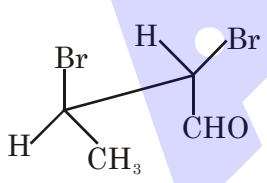


(A) Enantiomers (B) Diastereomers

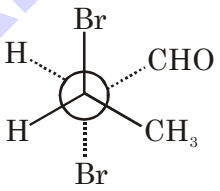
(C) Homomers (Identical) (D) Constitutional isomers

SE0056

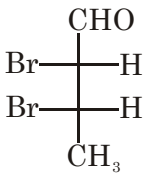
17. Identify the correct statement regarding following molecules?



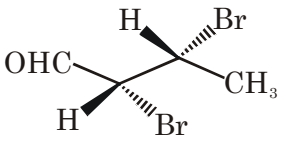
(M)



(N)



(O)



(P)

(A) **M** and **O** are diastereomers

(B) **N** and **P** are enantiomers

(C) **M** and **N** are identical

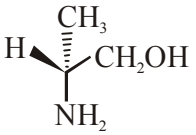
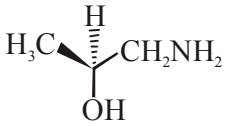
(D) **O** and **P** are diastereomers

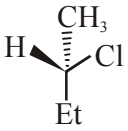
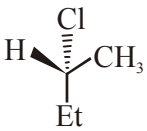
SE0057

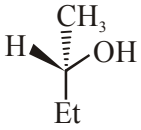
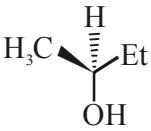
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

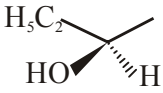
Matrix Match Type :

18. **Column I** **Column II**

(A)  &  (P) Structural

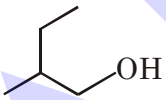
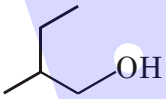
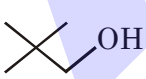
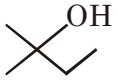
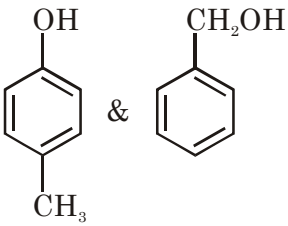
(B)  &  (Q) Identical

(C)  &  (R) Enantiomers

(D)  &  (S) Diastereomers

SE0058

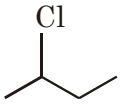
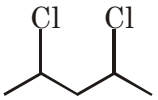
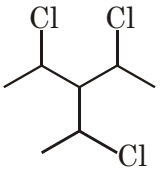
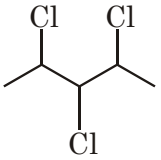
19. **Match the column :-**

	Column-I		Column-II
(1)	 & 	(P)	Position isomers
(2)	 & 	(Q)	Chain isomers
(3)	 & 	(R)	Homologues
(4)		(S)	Functional isomers

SE0059

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\65

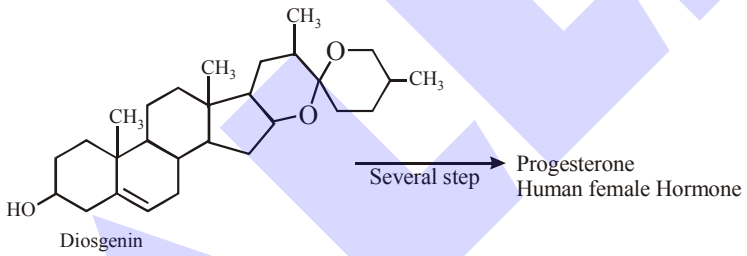
20. Match the column–I :-

Column-I (Compounds)		Column-II (Total number of stereoisomers)	
(1)		**(P)**	8
(2)		**(Q)**	4
(3)		**(R)**	3
(4)		**(S)**	2

SE0060

Subjective Type :

21.

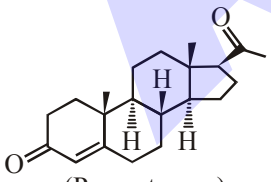


Diosgenin

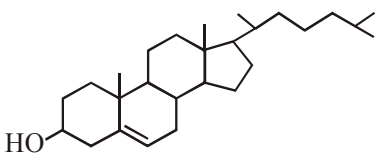
What is number of chiral centres present in Diosgenen is :

SE0061

22. Calculate the total number of chrial carbon atoms in.

(i) 

(Progesterone)

(ii) 

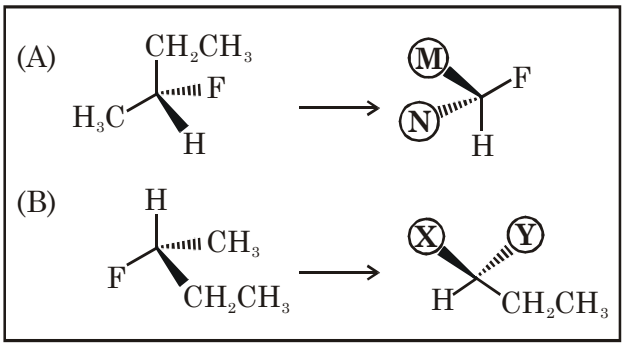
SE0062

23. Total number of isomeric (including stereo) bromochlorofluoroiodo propadiene.

SE0063

node06\B0BB-BCV\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_[OC]\Eng\04_Isomerism\01_Theory & Ex.p65

24. Re-orient the molecule at the left to match the partially drawn perspective at the right. Find the two missing substituents at their correct positions.



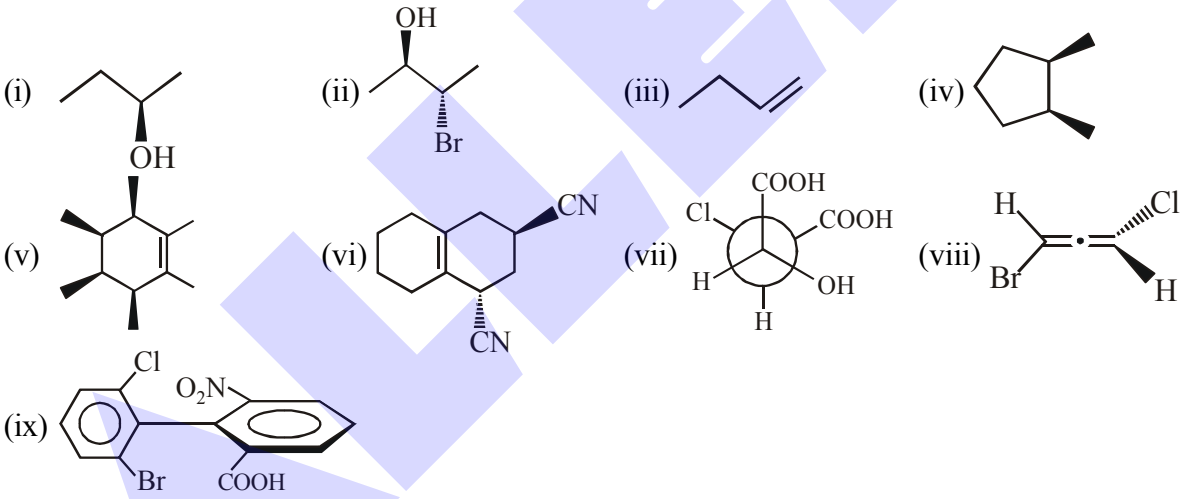
(A) **M** = CH_3CH_2- **N** = CH_3- (B) **X** = CH_3- **Y** = F – (C) **M** = CH_3- **N** = CH_3CH_2- (D) **X** = F – **Y** = CH_3-

SE0064

25. Find out the total number of cyclic isomers of C_6H_{12} which are optically active ?

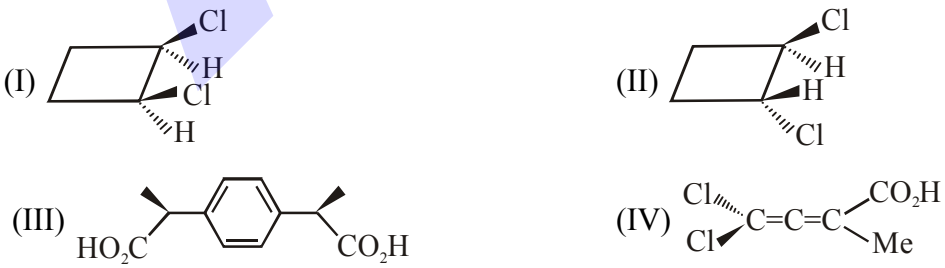
SE0065

26. How many of the given compounds are chiral :



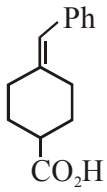
SE0066

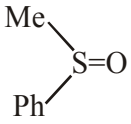
27. With reasons, state whether each of the following compounds I to VIII is chiral



node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

(V) 

(VI) 

(VII) 

SE0067

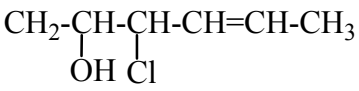
28. How many cyclopentane structures (including stereo) are possible for C_7H_{14}.

SE0068

29. The number of diastereoisomers (excluding enantiomers) for 1-bromo-2-chloro-3-iodocyclopropane.

SE0069

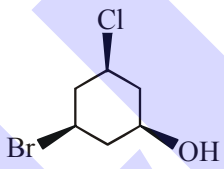
30. Identify total number of stereoisomers for the following compound :

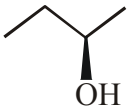


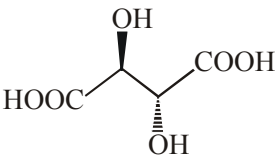
SE0070

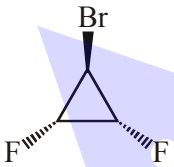
31. How many of the given molecule / species are chiral :

(I) 

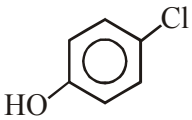
(II) 

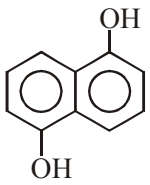
(III) 

(IV) 

(V) 

(VI) 

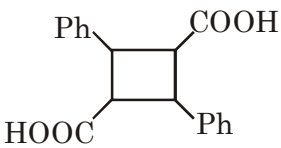
(VII) 

(VIII) 

(IX) CH_4

SE0071

32. Total geometrical isomers possible for :



(A) 3 (B) 4 (C) 5 (D) 8

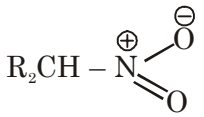
SE0072

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

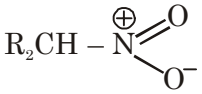
EXERCISE # S-I (STEREOISOMERISM)

Single Correct Type :

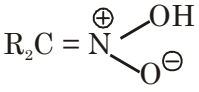
1. The correct statements describing the relationship between :



(X)



(Y)

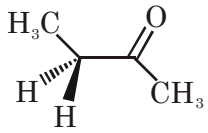


(Z)

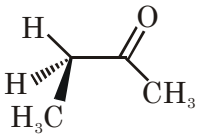
(A) **X** and **Y** are resonance structures and **Z** is a tautomer
(B) **X** and **Y** are tautomers and **Z** is resonance structure
(C) **X**, **Y** and **Z** are all resonance structures
(D) **X**, **Y** and **Z** all are tautomers

SE0073

2. The correct statements about conformations X and Y of 2-butanone are :



(X)



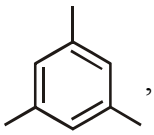
(Y)

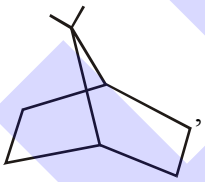
(a) **X** is more stable than **Y**
(b) **Y** is more stable than **X**
(c) Methyl groups in **X** are anti
(d) Methyl groups in **Y** are gauche

(A) a and d (B) a and c (C) b and c (D) a, c and d

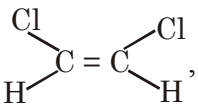
SE0074

3. Among the following, the number of molecules that possess C_2–axis of symmetry is :

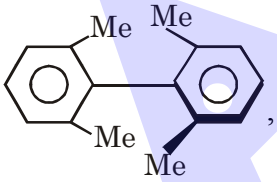
 ,

 ,

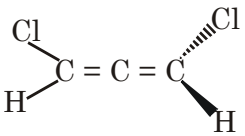
CH_4,

 ,

BF_3,

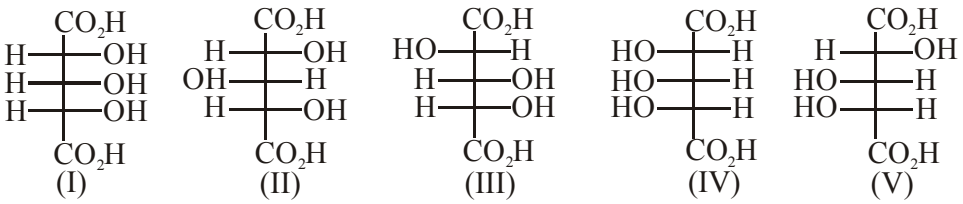
 ,

$CHCl_3$, 2,5-dimethylthiophene,



SE0075

4. Observe the given compounds and answer the following questions.



(I) (II) (III) (IV) (V)

(i) Which of the above formulae represent identical compounds ?

(A) I and II (B) I and IV (C) II and IV (D) III and IV

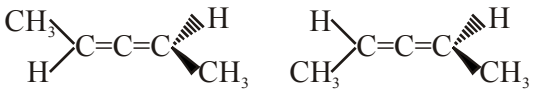
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

(ii) Which of the above compounds are enantiomers ?

(A) II and III (B) III and IV (C) III and V (D) I and V

SE0076

5. Which of the following option is correct regarding the given compounds :

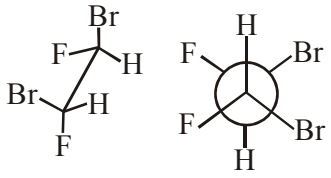


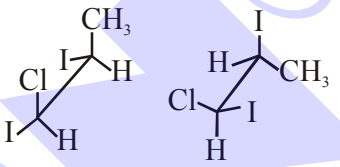
(A) Both are identical (B) Both are optically inactive

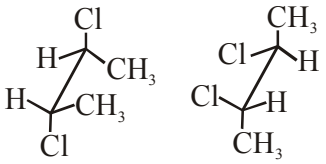
(C) Both are enantiomers (D) Geometrical isomer

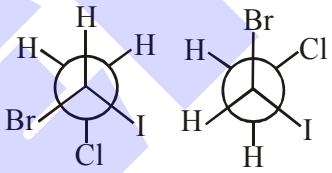
SE0077

6. Which of the following pairs of compound is/are identical ?

(A) 

(B) 

(C) 

(D) 

SE0078

Multiple Correct Type :

7. Which of the following statements is/are not correct for D-(+) glyceraldehyde :

(A) The symbol D indicates the dextrorotatory nature of the compound

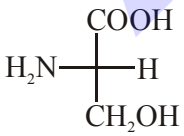
(B) The sign(+) indicates the dextrorotatory nature of the compound

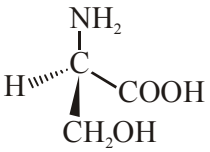
(C) The symbol D indicates that (–OH) group lies left to the chiral centre in the conventionally correct Fischer projection diagram

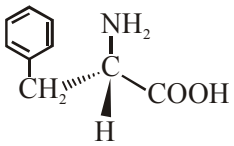
(D) The symbol D indicates that (–OH) group lies right to the chiral centre in the conventionally correct Fischer projection diagram

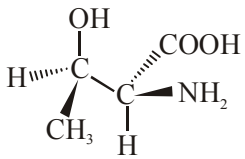
SE0079

8. Which of the following are correct representation of L-amino acids :

(A) 

(B) 

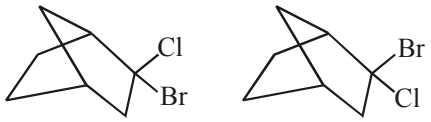
(C) 

(D) 

SE0080

node06\B0BB-BCV\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

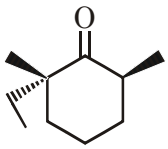
9. Identify relation between these two compounds :

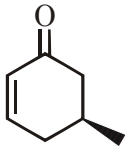


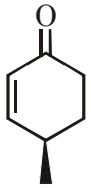
(A) Homomers (B) Enantiomers (C) Diastereomers (D) Positional Isomers

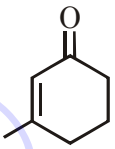
SE0081

10. Which of the following undergoes racemisation in alkaline medium?

(A) 

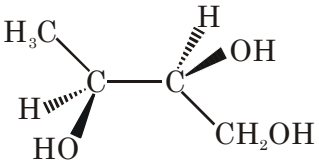
(B) 

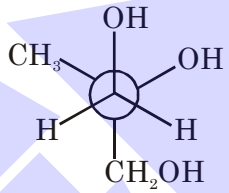
(C) 

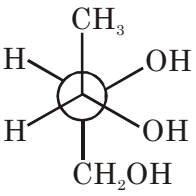
(D) 

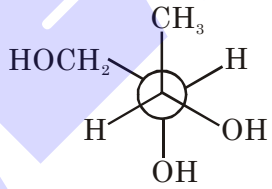
SE0082

11. Which compound is different from the others?

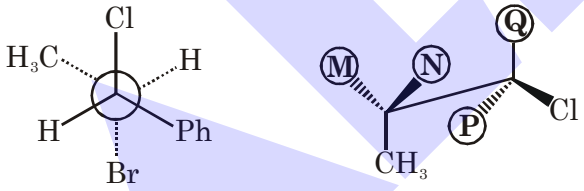
(A) 

(B) 

(C) 

(D) 

SE0083

12. 

(X) **(Y)**

What would be the correct match to get (**Y**) as a diastereomers of (**X**)?

(A) **M** = – H **Q** = – Ph
N = – Br **P** = –H

(B) **M** = – H **Q** = – H
N = – Br **P** = –Ph

(C) **M** = – Br **Q** = – Ph
N = – H **P** = –H

(D) **M** = – Br **Q** = – H
N = – H **P** = –Ph

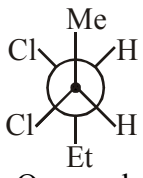
SE0084

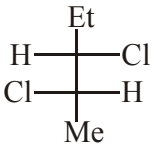
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

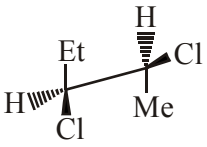
Comprehension Type :

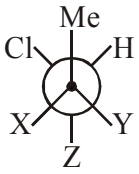
Paragraph for Question 13 and 14

Four compounds are given below 'S' is a stereoisomer of P.

(P) 

(Q) 

(R) 

(S) 

13. P & Q are related as :

(A) Identical (B) Enantiomer (C) Diastereomer (D)Positional isomerism

SE0085

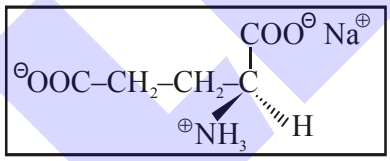
14. Which of the above structures represented is Sawhorse projection :-

(A) P (B) Q (C) R (D) S

SE0086

Paragraph for Question 15 to 17

S(+) Mono sodium Glutamate (MSG) is a flavour enhancer used in many foods. Fast foods often contain substantial amount of MSG and is widely used in Chinese food. If one mole of above MSG was placed in 845 ml solution and passed through 200 mm tube, the observed rotation was found to be + 9.6°.



15. Find out the specific rotation of (–) MSG :

(A) + 24° (B) +56.8° (C) –48° (D) None of these

SE0087

16. Find out the approximate percentage composition of (–) MSG in a mixture containing (+) MSG and (–) MSG whose specific optical rotation is –20° :

(A) 83.3% (B) 16.7% (C) 91.6% (D) 74%

SE0088

17. If 33.8 g of (+) MSG was put in 338 ml solution and was mixed with 16.9g of (–) MSG put in 169 ml solution and the final solution was passed through 400 mm tube. Find out observed rotation of the final solution :

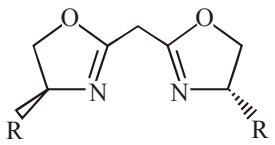
(A) +1.6° (B) +4.8° (C) +3.2° (D) None of these

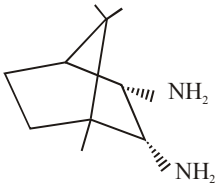
SE0089

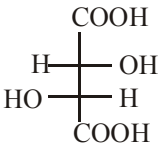
node06\B0BB-BCV\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_[OC]\Eng\04_Isomerism\01_Theory & Ex.p65

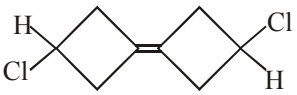
Paragraph for Q.18 and Q.19

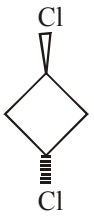
Among the following structures ?

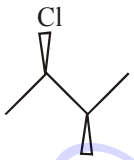
(I) 

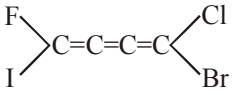
(II) 

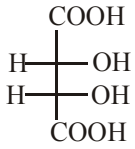
(III) 

(IV) 

(V) 

(VI) 

(VII) 

(VIII) 

18. Optically active compound is -

(A) III (B) IV (C) V (D) VII

SE0090

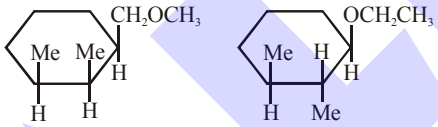
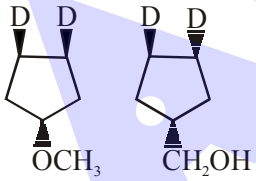
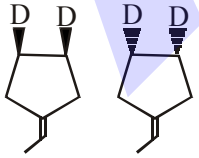
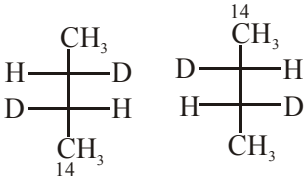
19. Which of the following will not show optical isomerism -

(A) I (B) II (C) V (D) VIII

SE0091

Matrix Match Type :

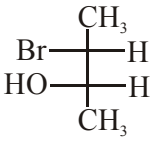
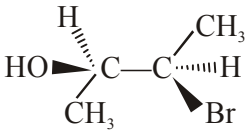
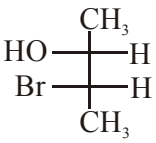
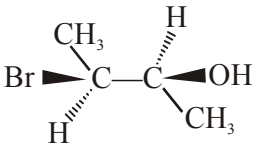
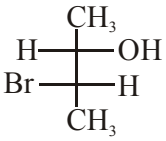
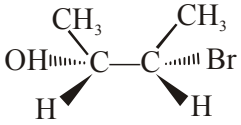
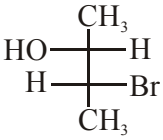
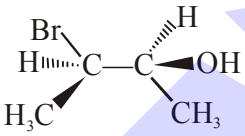
20.

Column-I (Compounds)	**Column-II (Relation)**
(A) 	(P) Metamers
(B) 	(Q) Functional Isomer
(C) 	(R) Geometrical isomer
(D) 	(S) Enantiomer
	(T) Diastereomer

SE0092

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\65

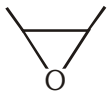
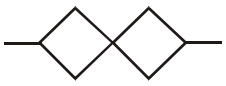
21. Match List-I, II, III with each other :

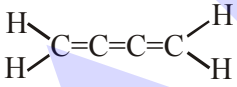
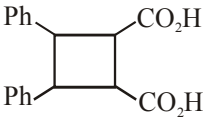
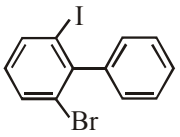
List - I	List - II	List - III
(A) 	(1) 	(i) (2R, 3R)
(B) 	(2) 	(ii) (2S,3S)
(C) 	(3) 	(iii) (2S,3R)
(D) 	(4) 	(iv) (2R,3S)

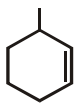
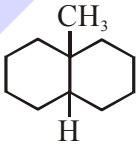
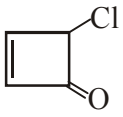
SE0093

Subjective Type :

22. In what stereoisomeric forms would you expect the following compounds to exist ?

(a) $EtCH(CO_2H)Me$ (b) $MeCH(CO_2Et)CO_2H$ (c)  (d) 

(e)  (f) Et(Me)C=C=C(Me)Et (g)  (h) 

(i)  (j)  (k) 

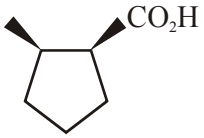
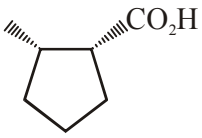
SE0094

23. Calculate the number of Benzenoid isomers possible for C_6H_3ClBrI.

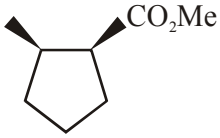
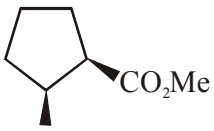
SE0095

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_[OC]\Eng\04_Isomerism\01_Theory & Ex.p65

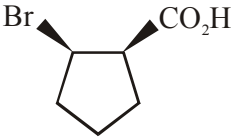
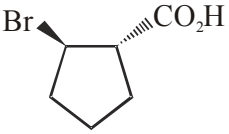
24. What are the relationships between the following pairs of isomers ?

(a)  and 

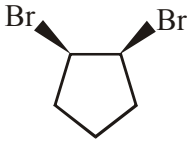
SE0096

(b)  and 

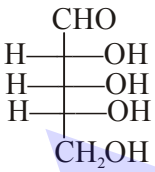
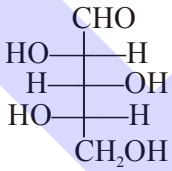
SE0096

(c)  and 

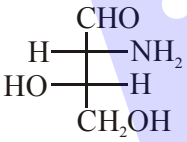
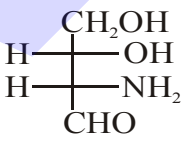
SE0096

(d)  and 

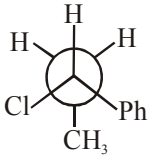
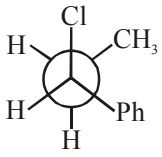
SE0096

(e)  and 

SE0097

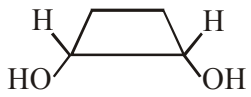
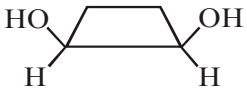
(f)  and 

SE0097

(g)  and 

SE0098

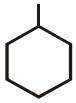
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

(h)  and 

SE0098

(i)  and 

SE0098

25. X $\xrightarrow[pt]{2H_2}$ 

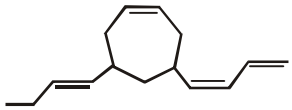
Find out total number of structures of X.

SE0099

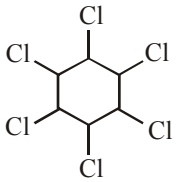
26. Calculate the number of chiral center in the molecule Ethyl 2,2-dibromo-4-ethyl-6-methoxy cyclohexane carboxylate.

SE0100

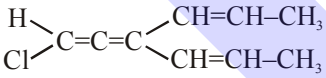
27. Calculate the total number of stereoisomers possible for

(i) 

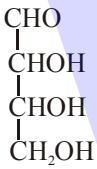
SE0101

(ii) 

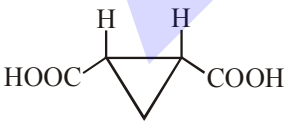
SE0102

(iii) 

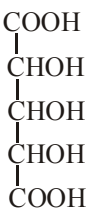
SE0103

(iv) 

SE0104

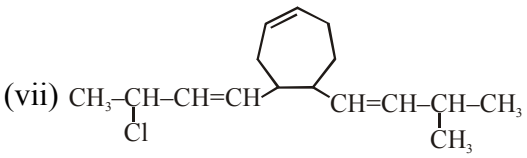
(v) 

SE0105

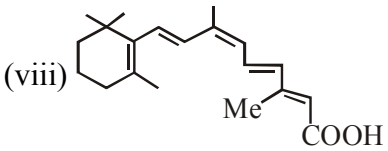
(vi) 

SE0105

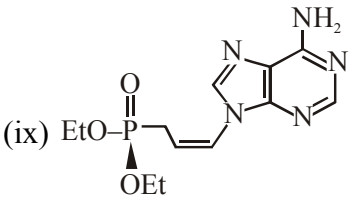
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

(vii) 

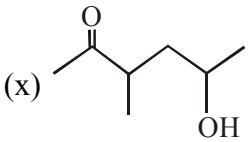
SE0105

(viii) 

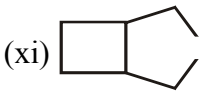
SE0105

(ix) 

SE0106

(x) 

SE0106

(xi) 

SE0106

28. How many different chloroethanes are there from the formula $C_2H_{6-n}Cl_n$ (where n can be any integer from 1 to 6)?

SE0107

ALLEN®

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\65

EXERCISE # J-MAINS

1. Geometrical isomerism is not shown by- **[AIEEE-2002]**

(1) 1,1–dichloro–1–pentene (2) 1,2–dichloro–1–pentene

(3) 1,3–dichloro–2–pentene (4) 1,4–dichloro–2–pentene

SR0048

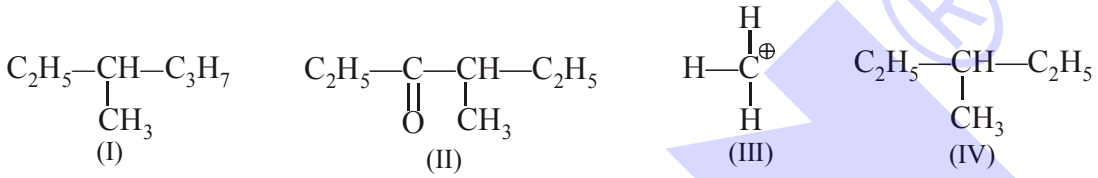
2. Racemic mixture is formed by mixing two : **[AIEEE-2002]**

(1) Isomeric compounds (2) Chiral compounds

(3) Meso compounds (4) Enantiomers with chiral carbon

SE0108

3. Among the following four structures I to IV **[AIEEE-2003]**



(I) (II) (III) (IV)

It is true that-

(1) All four are chiral compounds (2) Only I and II are chiral compounds

(3) Only III is a chiral compound (4) Only II and IV are chiral compounds

SE0110

4. Which of the following will have a meso-isomer also- **[AIEEE-2004]**

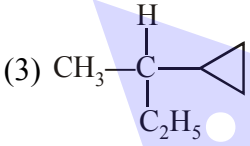
(1) 2–chlorobutane (2) 2,3–dichlorobutane

(3) 2,3–dichloropentene (4) 2–hydroxy propanoic acid

SE0111

5. Amongst the following compounds, the optically active alkane having lowest molecular mass is

(1) $CH_3—CH_2—C \equiv CH$ (2) $CH_3—CH_2—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_3$ **[AIEEE-2004]**

(3)  (4) $CH_3—CH_2—CH_2—CH_3$

SE0112

6. Which of following compounds is not chiral **[AIEEE-2005]**

(1) 1–chloropentane (2) 2–chloropentane

(3) 1–chloro–2–methyl pentane (4) 3–chloro–2–methyl pentane

SE0113

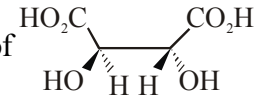
7. Increasing order of stability among the three main conformations (i.e. Eclipse, Anti, Gauche) of 2-fluoroethanol is **[AIEEE-2006]**

(1) Gauche, Eclipse, Anti (2) Eclipse, Anti, Gauche

(3) Anti, Gauche, Eclipse (4) Eclipse, Gauche, Anti

SR0049

node06\B0BB-BCV\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_[OC]\Eng\04_Isomerism\01_Theory & Ex.p65

8. The absolute configuration of  is : **[AIEEE-2008]**

(1) S, S (2) R, R (3) R, S (4) S, R

SE0116

9. The alkene that exhibits geometrical isomerism is :- **[AIEEE-2009]**

(1) 2–butene (2) 2–methyl–2–butene

(3) Propene (4) 2–methyl propene

SR0051

10. The number of stereoisomers possible for a compound of the molecular formula CH_3–CH=CH–CH(OH)–Me is :- **[AIEEE-2009]**

(1) 4 (2) 6 (3) 3 (4) 2

SE0117

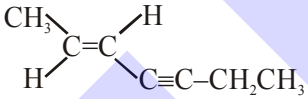
11. Out of the following, the alkene that exhibits optical isomerism is :- **[AIEEE-2010]**

(1) 2-methyl-2-pentene (2) 3-methyl-2-pentene

(3) 4-methyl-1-pentene (4) 3-methyl-1-pentene

SE0118

12. The IUPAC name of the following compounds is : **[JEE-MAIN-2012]**



(1) (Z) – 5 hepten – 3 – yne (2) (Z) – 2 hepten – 4 – yne

(3) (E) – 5 hepten – 3 – yne (4) (E) – 2 hepten – 4 – yne

SR0053

13. Dipole moment is shown by :- **[JEE-MAIN 2012]**

(1) trans-2, 3-dichloro- 2-butene (2) 1, 2-dichlorobenzene

(3) 1, 4-dichlorobenzene (4) trans-1, 2-dinitroethene

SR0054

14. Maleic acid and fumaric acids are :- **[JEE-MAIN 2012]**

(1) Tautomers (2) Chain isomers

(3) Geometrical isomers (4) Functional isomers

SR0055

15. Monocarboxylic acids are functional isomers of : **[JEE-MAIN 2013]**

(1) Esters (2) Amines

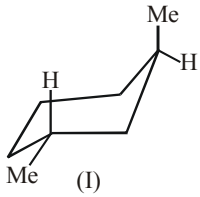
(3) Ethers (4) Alcohols

SR0056

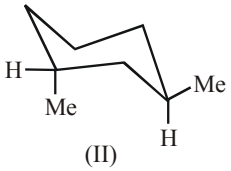
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\65

16. Arrange in the correct order of stability (decreasing order) for the following molecules:

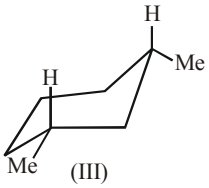
[JEE-MAIN 2013]



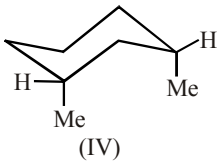
(I)



(II)



(III)



(IV)

(1) (I) > (II) > (III) > (IV)

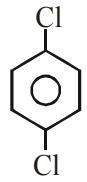
(2) (IV) > (III) > (II) ≈ (I)

(3) (III) > (I) ≈ (II) > (IV)

(4) (I) > (ll) ≈ (III) > (IV)

SR0057

17. For which of the following molecule significant $\mu \neq 0$ **[JEE-MAIN 2014]**

(I) 

(II) 

(III) 

(IV) 

(1) Only (III)

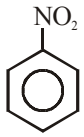
(2) (III) and (IV)

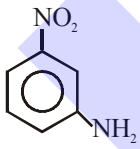
(3) Only (I)

(4) (I) and (II)

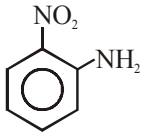
SR0058

18. Which compound exhibits maximum dipole moment among the following :-**[JEE-MAIN 2015]**

(1) 

(2) 

(3) 

(4) 

SR0059

19. The number of structural isomers for C_6H_{14} is: **[JEE-MAIN 2015]**

(1) 6 (2) 4 (3) 3 (4) 5

SR0060

20. The optically inactive compound from the following is :- **[JEE-MAIN-2015]**

(1) 2-chloropropanal

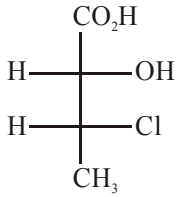
(2) 2-chlorobutane

(3) 2-chloro-2-methylbutane

(4) 2-chloropentane

SE0119

21. The absolute configuration of : **[JEE-MAIN-2016]**



(1) (2R, 3R)

(2) (2R, 3S)

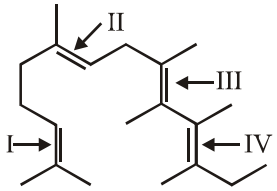
(3) (2S, 3R)

(4) (2S, 3S)

SE0120

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

22. In the following structure, the double bonds are marked as I, II, III and IV **[JEE-MAIN-2017]**



Geometrical isomerism is **not** possible at site (s)

(1) III (2) I (3) III and IV (4) I and III

SE0121

23. Which of the following compounds will show the maximum enol content? **[JEE-MAIN-April 2019]**

(1) $CH_3COCH_2COCH_3$ (2) CH_3COCH_3

(3) $CH_3COCH_2CONH_2$ (4) $CH_3COCH_2COOC_2H_5$

SE0122

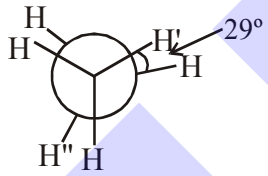
24. Which of these factors does not govern the stability of a conformation in acyclic compounds ? **[JEE-MAIN-April 2019]**

(1) Torsional strain (2) Angle strain

(3) Steric interactions (4) Electrostatic forces of interaction

SE0123

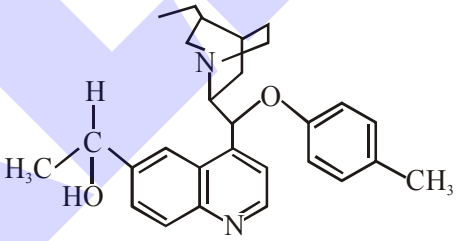
25. In the following skew conformation of ethane, H'–C–C–H" dihedral angle is : **[JEE-MAIN-April 2019]**



(1) 120° (2) 58° (3) 149° (4) 151°

SE0124

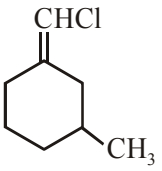
26. The number of chiral carbons present in the molecule given below is _____ . **[JEE-MAIN 2020]**

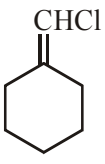


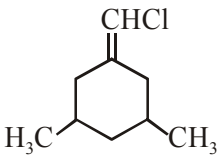
SE0143

27. Among the following compounds, geometrical isomerism is exhibited by : **[JEE-MAIN 2020]**

(1) 

(2) 

(3) 

(4) 

SE0144

28. Which of the following compound shows geometrical isomerism **[JEE-MAIN 2020]**

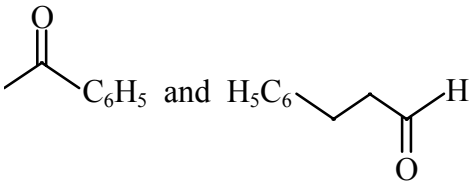
(1) 2-methylpent-2-ene (2) 4-methylpent-l-ene (3) 4-methylpent-2-ene (4) 2-methylpent-l-ene

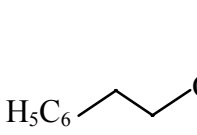
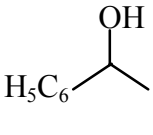
SE0145

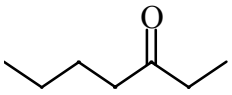
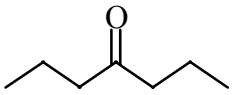
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\65

29. Which one of the following pairs of isomers is an example of metamerism ? **[JEE-MAIN 2021]**

(1) $CH_3CH_2CH_2CH_2CH_3$ and $H_3C-\underset{\displaystyle CH_3}{\overset{\displaystyle CH}{\underset{|}{\overset{|}{C}}}}-CH_3$

(2)  and

(3)  and 

(4)  and 

SE0146

30. Which of the following molecules does not show stereo isomerism ? **[JEE-MAIN 2021]**

(1) 3,4-Dimethylhex-3-ene (2) 3-Methylhex-1-ene

(3) 3-Ethylhex-3-ene (4) 4-Methylhex-1-ene

SE0147

31. The number of acyclic structural isomers (including geometrical isomers) for pentene are ____ **[JEE-MAIN 2021]**

SE0148

32. The dihedral angle in staggered form of Newman projection of 1, 1, 1-Trichloro ethane is ..... degree. (Round off to the nearest integer) **[JEE-MAIN 2021]**

SE0149

33. Staggered and eclipsed conformers of ethane are : **[JEE-MAIN 2021]**

(1) Polymers (2) Rotamers (3) Enantiomers (4) Mirror images

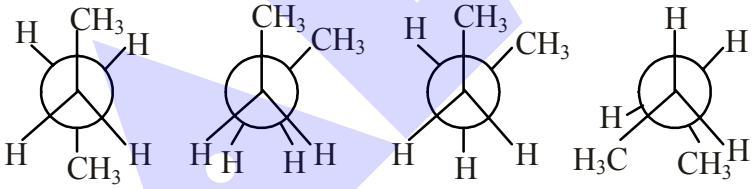
SE0150

34. The number of stereoisomers possible for 1,2-dimethyl cyclopropane is : **[JEE-MAIN 2021]**

(1) One (2) Four (3) Two (4) Three

SE0151

35. Arrange the following conformational isomers of n-butane in order of their increasing potential energy :



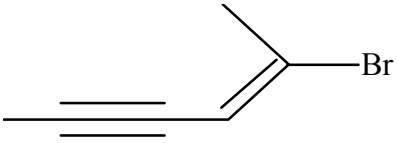
I II III IV

[JEE-MAIN 2021]

(1) II < III < IV < I (2) I < IV < III < II

(3) II < IV < III < I (4) I < III < IV < II **SE0152**

36. Choose the correct name for compound given below : **[JEE-MAIN 2021]**



(1) (4E)-5-Bromo-hex-4-en-2-yne (2) (2E)-2-Bromo-hex-4-yn-2-ene

(3) (2E)-2-Bromo-hex-2-en-4-yne (4) (4E)-5-Bromo-hex-2-en-4-yne

SE0153

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

EXERCISE # J-ADVANCED (Objective)

1. Which of the following compound will exhibits geometrical isomerism : **[IIT-2000]**

(A) 1–phenyl–2–butene (B) 3–phenyl–1–butene

(C) 2–phenyl–1–butene (D) 1, 1–diphenyl–1–propene

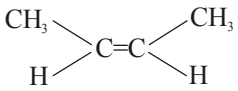
SR0065

2. The number of isomers for the compound with molecular formula $C_2BrClFI$ is : **[IIT-2000]**

(A) 3 (B) 4 (C) 5 (D) 6

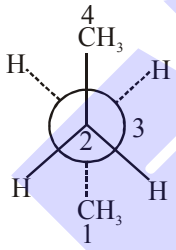
SR0066

3. Which of the following has the lowest dipole moment : **[IIT-2000]**

(A)  (B) $CH_3–C\equiv C–CH_3$ (C) $CH_3CH_2C\equiv CH$ (D) $CH_2=CH–C\equiv CH$

SR0067

4. In the given conformation, If C_2 is rotated about $C_2–C_3$ bond anticlockwise by an angle of 120° then the conformation obtained is : **[IIT-2004]**



(A) Fully eclipsed conformation (B) Partially eclipsed conformation

(C) Gauche conformation (D) Staggered conformation

SR0068

5. (i) $\mu_{obs} = \sum_i \mu_i x_i$

where μ_i is the dipole moment of a stable conformer of the molecule, $Z–CH_2–CH_2–Z$ and x_i is the mole fraction of the stable conformer. **[IIT- 2005]**

Given : μ_{obs} = 1.0 D and x (Anti) = 0.82

Draw all the stable conformers of $Z–CH_2–CH_2–Z$ and calculate the value of $\mu_{(Gauche)}$.

(ii) Draw the stable conformer of Y–CHD–CHD–Y(meso form), when Y = CH_3 (rotation about $C_2–C_3$) and Y = OH (rotation about $C_1–C_2$) in Newmann projection.

SR0069

6. **Statement-I :** Molecules that are not superimposable on their mirror images are chiral

Because

Statement-II : All chiral molecules have chiral centres. **[IIT-2007]**

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1

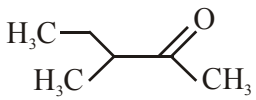
(C) Statement-1 is True, Statement-2 is False.

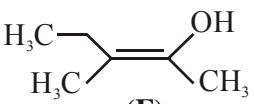
(D) Statement-1 is False, Statement-2 is True.

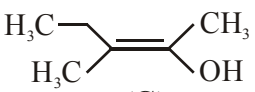
SE0131

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\05

7. The correct statement(s) concerning the structures E, F and G is (are) **[IIT-2008]**


(E)


(F)

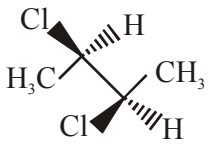

(G)

(A) **E, F** and **G** are resonance structures (B) **E, F** and **E, G** are tautomers

(C) **F** and **G** are geometrical isomers (D) **F** and **G** are diastereomers

SE0132

8. The correct statement(s) about the compound given below is (are) : **[IIT-2008]**



(A) The compound is optically active

(B) The compound possesses centre of symmetry

(C) The compound possesses plane of symmetry

(D) The compound possesses axis of symmetry

SE0133

9. The correct statement(s) about the compound $H_3C(HO)HC-CH=CH-CH(OH)CH_3$ **(X)** is (are) :

(A) The total number of stereoisomers possible for X is 6 **[IIT-2009]**

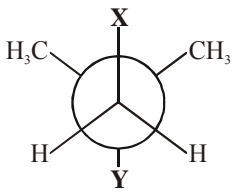
(B) The total number of diastereomers possible for X is 3

(C) If the stereochemistry about the double bond in X is trans, the number of enantiomers possible for X is 4

(D) If the stereochemistry about the double bond in X is cis, the number of enantiomers possible for X is 2

SE0134

10. In the Newman projection for 2,2–dimethylbutane **[IIT-2010]**



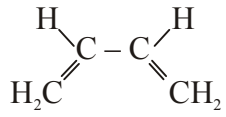
X and **Y** can respectively be –

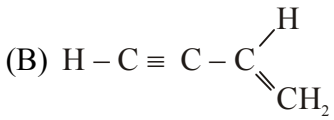
(A) H and H (B) H and C_2H_5 (C) C_2H_5 and H (D) CH_3 and CH_3

SE0135

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\02_Isomerism\01_Theory & Ex.p65

11. Amongst the given option, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are) - **[IIT-2011]**

(A) 

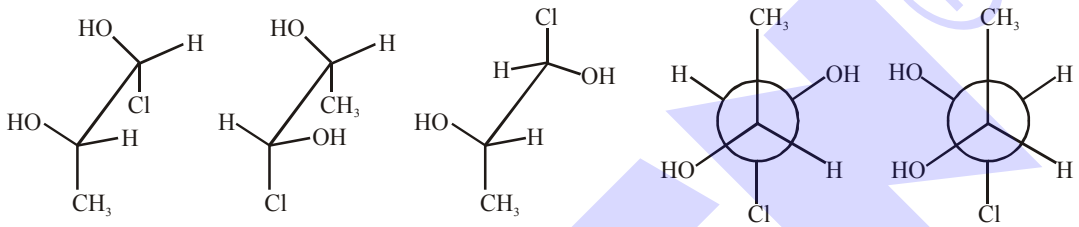
(B) 

(C) $H_2C = C = O$

(D) $H_2C = C = CH_2$

SR0072

12. Which of the given statement(s) about N,O,P and Q with respect to M is (are) correct ? **[JEE-2012]**



M **N** **O** **P** **Q**

(A) M and N are non-mirror image stereoisomers

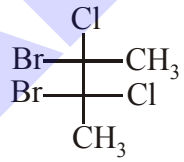
(B) M and O are identical

(C) M and P are enantiomers

(D) M and Q are identical

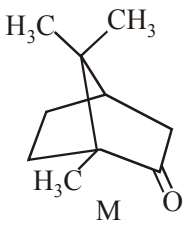
SE0136

13. The total number(s) of **stable** conformers with **non-zero** dipole moment for the following compound is (are) **[JEE-2014]**



SE0137

14. The total number of stereoisomers that can exist for M is : **[JEE-2015]**

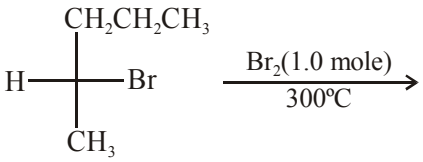


M

SE0138

node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. displ, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp05

15. In the following monobromination reaction, the number of possible chiral products is : **[JEE-2016]**



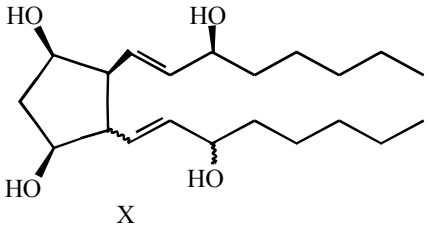
(1.0 mole)

(enatiomerically pure)

SE0139

16. For the given compound X, the total number of optically active stereoisomers is____.

[IIT-JEE 2018]



X

This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is fixed

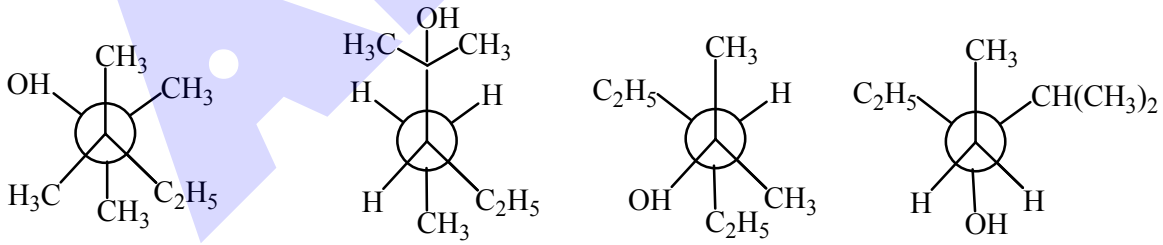
This type of bond indicates that the configuration at the specific carbon and the geometry of the double bond is NOT fixed

SE0140

17. Total number of isomers, considering both structural and stereoisomers, of cyclic ethers with the molecular formula C_4H_8O is ___ **[IIT-JEE 2019]**

SE0141

18. Newman projections **P**, **Q**, **R** and **S** are shown below : **[IIT-JEE 2020]**



P **Q** **R** **S**

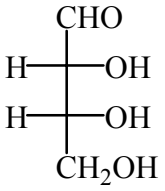
Which one of the following options represents identical molecules ?

(A) P and Q (B) Q and S (C) Q and R (D) R and S

SE0154

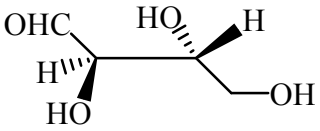
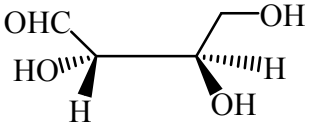
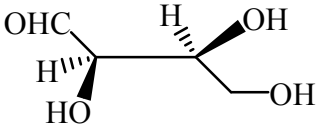
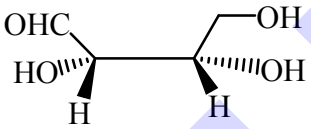
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Nome, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

19. The Fischer projection of D-erythrose is shown below. **[IIT-JEE 2020]**



D-Erythrose

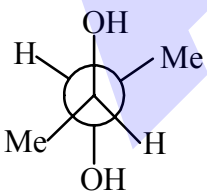
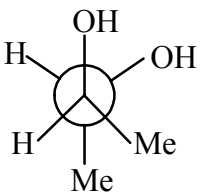
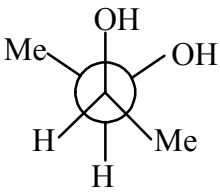
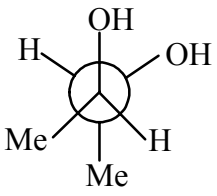
D-Erythrose and its isomers are listed as **P**, **Q**, **R** and **S** in **Column-I**. Choose the correct relationship of **P**, **Q**, **R** and **S** with D-erythrose from **Column II**.

Column-I		**Column-II**	
(P)		**1.**	Diastereomer
(Q)		**2.**	Identical
(R)		**3.**	Enantiomer
(S)			

(A) P→2, Q→3, R→2, S→2
(B) P→3, Q→1, R→1, S→2
(C) P→2, Q→1, R→1, S→3
(D) P→2, Q→3, R→3, S→1

SE0155

20. Among the following, the conformation that corresponds to the most stable conformation of meso-butane-2,3-diol is - **[IIT-JEE 2021]**

(A) 
(B) 
(C) 
(D) 

SE0156

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ANSWER KEY

EXERCISE # O-I (MAINS ORIENTED)

Single Correct Type :

1. Ans. (D)	**2.** Ans. (A)	**3.** Ans. (C)	**4.** Ans. (C)	**5.** Ans. (C)
6. Ans. (C)	**7.** Ans. (D)	**8.** Ans. (D)	**9.** Ans. (A)	**10.** Ans. (B)
11. Ans. (C)	**12.** Ans. (A)	**13.** Ans. (C)	**14.** Ans. (B)	**15.** Ans. (A)
16. Ans. (D)	**17.** Ans. (B)	**18.** Ans. (D)	**19.** Ans. (C)	**20.** Ans. (D)
21. Ans. (A)	**22.** Ans. (B)	**23.** Ans. (D)	**24.** Ans. (B)	**25.** Ans. (A)
26. Ans. (B)	**27.** Ans. (C)			

More than one correct Type :

28. Ans. (B,C,D) **29.** Ans. (A,C,D) **30.** Ans. (A,B,D) **31.** Ans. (B,C,D)

Assertion / Reasoning Type :

32. Ans. (D) **33.** Ans. (C)

Comprehension Type :

34. Ans. (C) **35.** Ans. (D)

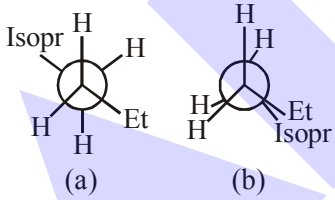
Matrix Match Type :

36. Ans. (A)→P ; (B)→R ; (C)→Q ; (D)→R

37. Ans. (A)→R ; (B)→S ; (C)→P ; (D)→Q

Subjective Type :

38. Ans.



39. Ans. (a) cis (b) cis (c) cis (d) trans (e) trans (f) trans

40. Ans.



41. Ans. (4) **42.** Ans. II, III, IV, V, VI, VIII, IX, XI, XIII, XIV,

43. Ans. (10) **44.** Ans. (2112) **45.** (a) Ans. (4) ; (b) Ans. (0404)

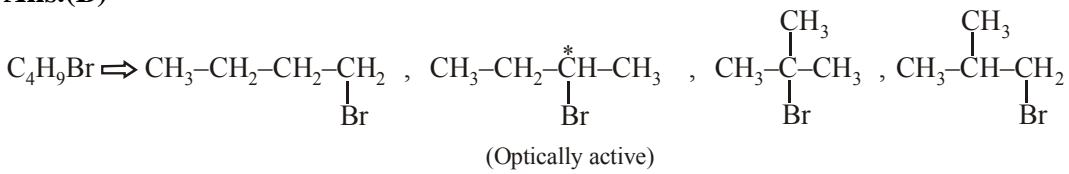
46. Ans. (a) Position isomers so (3) (b) Geometrical isomers so (1)

(c) Functional isomers so (4) (d) Chain isomers so (2)

47. Ans.(C)	**48.** Ans.(C)	**49.** Ans.(D)	**50.** Ans.(D)
51. Ans.(A)	**52.** Ans.(D)	**53.** Ans.(A)	**54.** Ans.(D)

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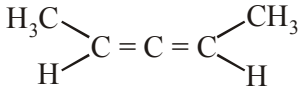
55. Ans.(B)



Optically active isomers ⇒ 2

56. Ans.(B) **57. Ans.(D)** **58. Ans.(A)**

59. Ans.(C)



60. Ans.(B) **61. Ans.(C)** **62. Ans.(D)** **63. Ans.(C)**

64. Ans.(A)

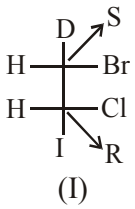
Cl–CH=C=C=CH–Cl is a planar structure

65. Ans.(A)

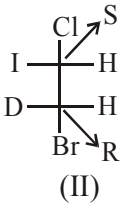
66. Ans.(C)

Meso tartaric acid and d-fartaric acid and not mirror image of each other so they are diastereomers.

67. Ans.(A)


(I)

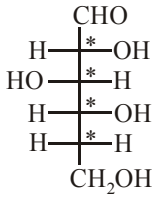
and


(II)

Enantiomers (I & II are mirror image of each other)

68. Ans.(C) **69. Ans.(D)** **70. Ans.(B)**

71. Ans.(C)



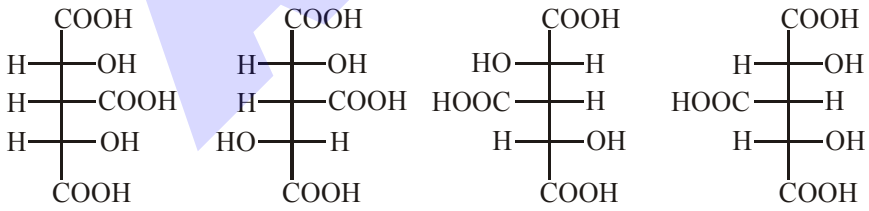
$2^n \Rightarrow 2^4 \Rightarrow 16$

72. Ans.(B) **73. Ans.(B)**

74. Ans.(B)

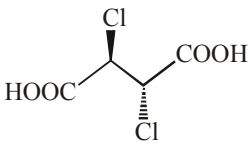
Optically active compounds are resolvable.

75. Ans.(B)



76. Ans.(C)

77. Ans.(A)

POS is present in 

78. Ans.(C)

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79. Ans.(B)

e.e = 87.5 – 12.5 $\Rightarrow$ 75 % of

$$ee = \frac{\text{Rotation by mixture}}{\text{Rotation by pure isomer}} \times 100$$

$$75 = \frac{X}{30} \times 100$$

$$X = \frac{75 \times 30}{100} = +22.5^{\circ}$$

80. Ans.(B)

ℓ = 10 dm

c = 20 gm/200 ml

$\alpha = 30^{\circ}$

$$[\alpha]_{specific} = \frac{\alpha_{obs}}{c.l} = \frac{30}{\frac{20}{200} \times 10} \Rightarrow 30^{\circ}$$

$\alpha_{obs.}$ after dilution

$\alpha_{obs} = \alpha_{sp}$ c.p.

$$\alpha_{obs} = 30 \times \frac{20}{1000} \times 10 \Rightarrow 6^{\circ}$$

81. Ans.(A)

$$\% \text{ optical purity} = \frac{|d - \ell|}{d + \ell} \times 100 = \frac{4}{8} \times 100 = 50\%$$

82. Ans.(C) **83. Ans.(A)** **84. Ans.(D)** **85. Ans.(C)** **86. Ans.(A)**

EXERCISE-O-II

Single correct Option Type :

1. Ans.(B)

$C_5H_{10}O$:

Ketones

(i) $CH_3–CH_2–CH_2–CH_2–\underset{\underset{O}{\|}}{C}–H$

(ii) $CH_3–CH_2–\underset{*}{\overset{\overset{CH_3}{|}}{CH}}–\underset{\underset{O}{\|}}{C}–H$ Optical active (Two)

(iii) $CH_3–\underset{\underset{CH_3}{|}}{CH}–CH_2–\underset{\underset{O}{\|}}{C}–H$

Total $\Rightarrow$ 5 aldehyde

(i) $CH_3–\overset{\overset{O}{\|}}{C}–CH_2–CH_2–CH_3$

(ii) $CH_3–CH_2–\overset{\overset{O}{\|}}{C}–CH_2–CH_3$

(iii) $CH_3–\overset{\overset{O}{\|}}{C}–\underset{\underset{CH_3}{|}}{CH}–CH_3$

(iv) $CH_3–\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}\!\!-\!\!-\underset{\underset{O}{\|}}{C}–H$

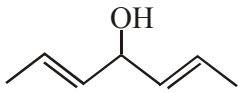
Total $\Rightarrow$ 3 ketones

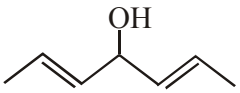
2. Ans.(C)

3. Ans.(B)

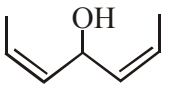
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp\, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Ex.p65

4. Ans.(A)

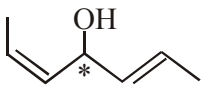


$\Rightarrow$ (i) 

(Trans, Trans)

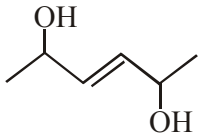
(ii) 

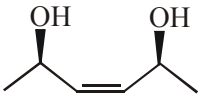
(Cis , Cis)

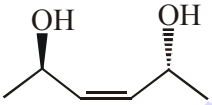
(iii) 

(Cis , Trans) Two $\Rightarrow$ R + S

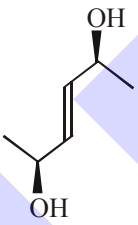
Total is isomers = 4



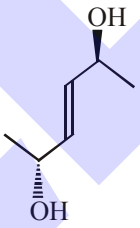
$\Rightarrow$ (i) 

(ii) 

(±) Two

(iii) 

(±) Two

(iv) 

Total = 6 isomer

5. Ans.(A,C,D)

6. Ans.(A,C)

Optically active compounds are resolvable and A & C are optically inactive

7. Ans.(B,C,D) **8. Ans.(A,B,C)** **9. Ans.(A,B,D)**

10. Ans.(C,D) **11. Ans.(A,D)** **12. Ans.(A,C)**

13. Ans.(A,B,C,D) **14. Ans.(C)** **15. Ans.(D)**

16. Ans.(C) **17. Ans.(D)**

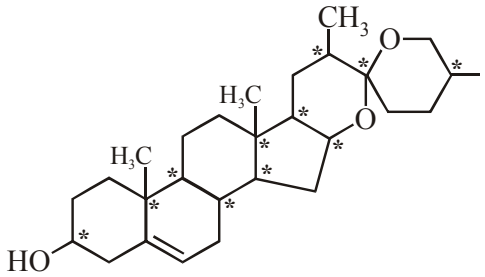
18. Ans.(A)→P ; (B)→R ; (C)→Q ; (D)→R

19. Ans.(1→R, 2→P, 3→Q, 4→S)

20. Ans.(1→S, 2→R, 3→Q, 4→Q)

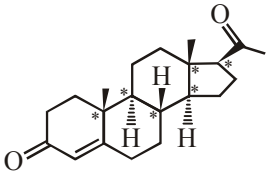
node06\B0BB-BC\Kota\JEE(Advanced)\Nurture\Chem\Module\Module#Name, Ele. disp, A & B Strength, Isom, Purification_(OC)\Eng\04_Isomerism\01_Theory & Exp\65

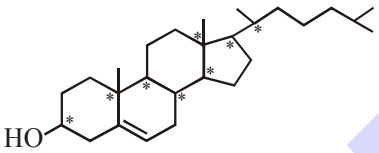
21. Ans.(11)



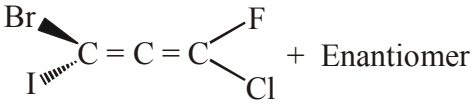
22. Ans. (i) 6, (ii) 8

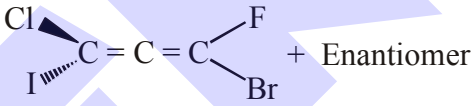
Explanation

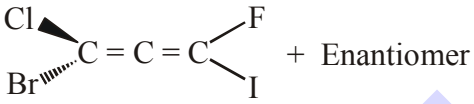
(i) 

(ii) 

23. Ans.(6)

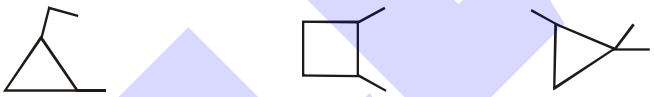
 + Enantiomer

 + Enantiomer

 + Enantiomer

24. Ans.(A,B)

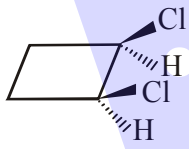
25. Ans.(8)

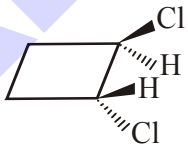


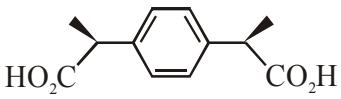
4 Optically Active Isomers 2 Optically Active 2 Optically Active

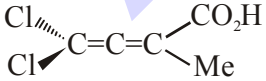
2 + 2 + 4 = 8

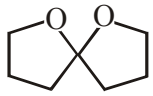
26. Ans.(6)

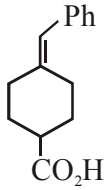
(I) 

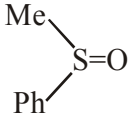
(II) 

(III) 

(IV) 

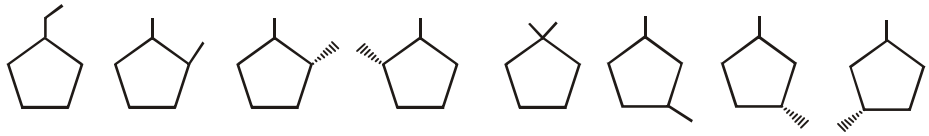
(V) 

(VI) 

(VII) 

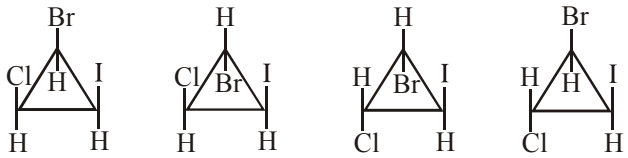
27. Ans. Achiral : I, III , IV ; Chiral : II,V, VI, VII

28. Ans.(8)

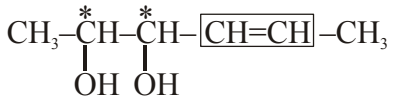


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29. Ans.(4)



30. Ans.(8)



Stereogenic centre = 3

Total number of stereoisomer = $2^3 = 8$

31. Ans.(3)

32. Ans.(C)

EXERCISE # S-I

Single Correct Type :

1. Ans.(A) **2. Ans.(D)** **3. Ans.(8)**

4. Ans.(i)-(B) ; (ii)-(C) **5. Ans.(C)** **6. Ans.(C)**

Multiple Correct Type :

7. Ans.(A,C) **8. Ans.(A,C,D)** **9. Ans.(C)**

10. Ans.(C) **11. Ans.(B)** **12. Ans.(A,D)**

Comprehension Type :

13. Ans.(B) **14. Ans.(C)**

15. Ans.(D)

M.W. of MSG = 169

$$C = \frac{169 \text{ gm}}{845 \text{ ml}}$$

$\ell = 200 \text{ mm} = 2 \text{ dm}$

$\alpha_{obs} = +9.6^\circ$

$$[\alpha]_{sp} = \frac{\alpha_{obs}}{C.\ell.} = \frac{9.6}{\frac{169}{845} \times 2} = -24^\circ$$

16. Ans.(C)

$$ee = \frac{[\alpha]_{mixture}}{[\alpha]_{pure}} \times 100 = \frac{-20^\circ}{-24^\circ} \times 100 = 83.3^\circ$$

$\therefore$ RM = 100 – 83.3 $\Rightarrow$ 16.7 % → d = 8.35, l = 8.35

Total (–) MSG = 83.3 + 8.35

= 91.6 %

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17. Ans.(C)

$$C = \frac{33.8-16.9\ g}{338+169\ ml} = \frac{16.9\ g}{507\ ml}$$

ℓ = 400 mm = 4 dm

$$\alpha_{obs} = [\alpha]_{sp.}\ c.\ell = 24\times\frac{16.9}{507}\times 4$$

$= +3.2^{\circ}$

18. Ans.(A) **19. Ans.(C)**

Matrix Match Type :

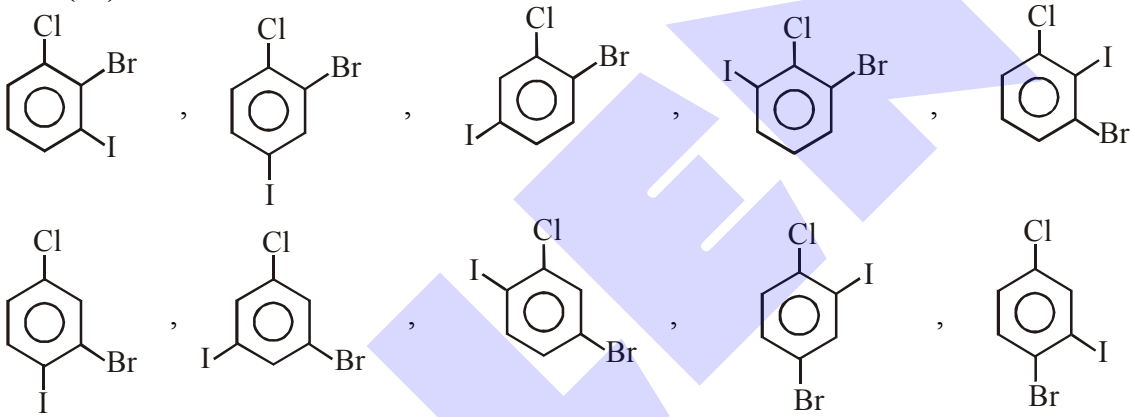
20. Ans.(A)→P ; (B)→Q ; (C)→R,S ; (D)→S

21. Ans.(A)→4→iii ; (B)→3→iv ; (C)→2→ii ; (D)→1→i

Subjective Type :

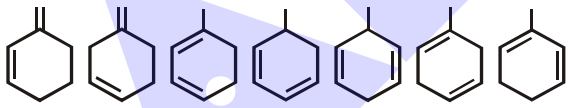
22. Ans. Optical : a, b, c, d, f, g, i, j, k ; Geometrical isomer : c, g, j ; None : e, h

23. Ans.(10)



24. Ans. (a) Enantiomers, (b) Enantiomers, (c) Geometrical isomers & Diastereomers,
(d) Positional, (e) Optical, (Diastereomers), (f) Diastereomers,
(g) Enantiomers, (h) Identical, (i) Geometrical isomers (Diastereomers)

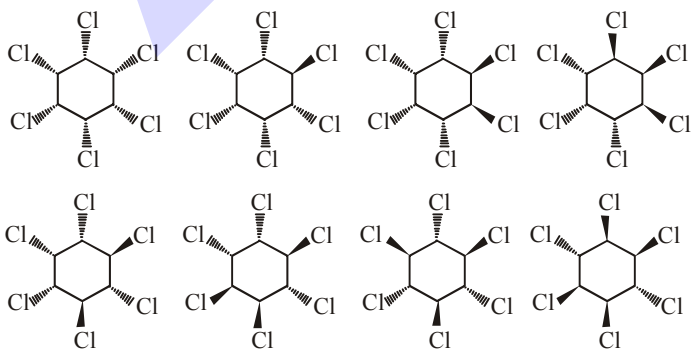
25. Ans.(7)



26. Ans.(3)

27. Ans.(i) 2^4 (ii) 9 (iii) 4 (iv) 4 (v) 3, (vi) 4 (vii) 2^5, (viii) 2^4, (ix) 2, (x) 4, (xi) 3

(ii)



28. Ans.(9)

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EXERCISE # J-MAINS

1. **Ans. (1)** **2.** **Ans. (4)** **3.** **Ans. (2)** **4.** **Ans. (2)**

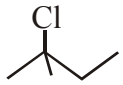
5. **Ans. (3)** **6.** **Ans. (1)** **7.** **Ans. (2)** **8.** **Ans. (2)**

9. **Ans. (1)** **10.** **Ans. (1)** **11.** **Ans. (4)** **12.** **Ans. (4)**

13. **Ans. (2)** **14.** **Ans. (3)** **15.** **Ans. (1)** **16.** **Ans. (3)**

17. **Ans. (2)** **18.** **Ans. (3)** **19.** **Ans. (4)**

20. **Ans. (3)**

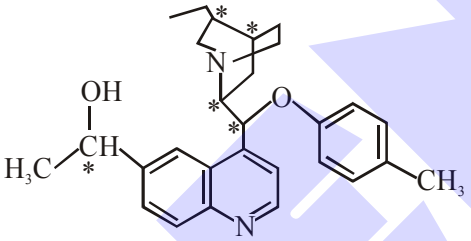
Sol. 

It achiral \ optically inactive

21. **Ans. (3)** **22.** **Ans. (2)** **23.** **Ans. (1)** **24.** **Ans. (2)** **25.** **Ans. (3)**

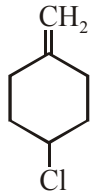
26. **Ans. (5.00)**

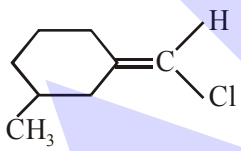
Sol. No. of chiral centres

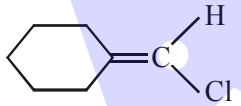


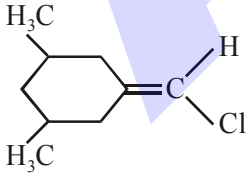
27. **NTA Answer (2)**

ALLEN Answer (2), (4)

Sol. (A)  Not show GI

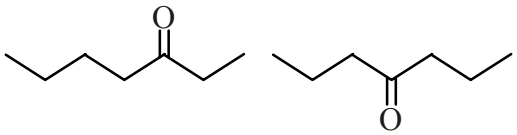
(2)  Show GI

(3)  Not show GI

(4)  Show GI

28. **Ans. (3)**

29. **Ans. (4)**

Sol. 

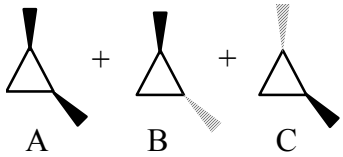
are metamers.

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30. Ans. (3) **31. Ans. (6)** **32. Ans. (60)** **33. Ans. (2)**

34. Ans. (4)

Sol. 

35. Ans. (4)

Sol. More stable less potential energy.
Stability order : I > III > IV > II
So
Potential energy : II > IV > III > I

36. Ans. (3)

EXERCISE # (J-ADVANCED Objective)

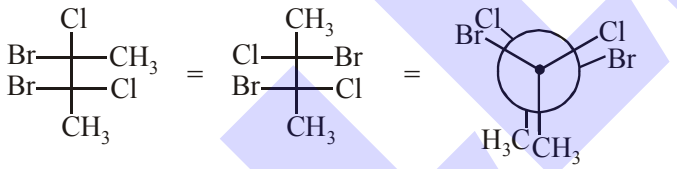
1. Ans. (A) **2. Ans. (D)** **3. Ans. (B)** **4. Ans. (C)**

5. Ans. (i) $\frac{1}{0.18}$D **, (ii) Anti form when Y = CH_3 & Gauche when Y= –OH**

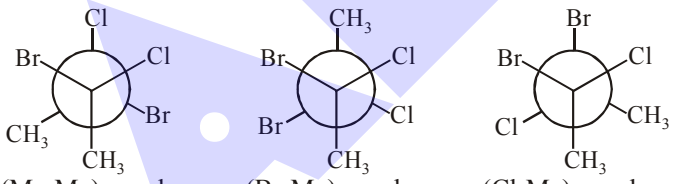
6. Ans. (C) **7. Ans. (B,C,D)** **8. Ans. (A,D)** **9. Ans. (A,D)**

10. Ans. (B,D) **11. Ans. (B,C)** **12. Ans. (A,B,C)**

13. Ans. (3)



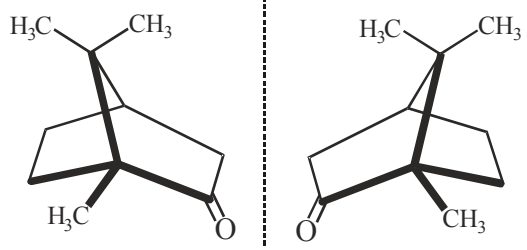
Stable conformer (with $\mu \neq 0$)



(Me-Me) gauche (Br-Me) gauche (Cl-Me) gauche

14. Ans. (2)

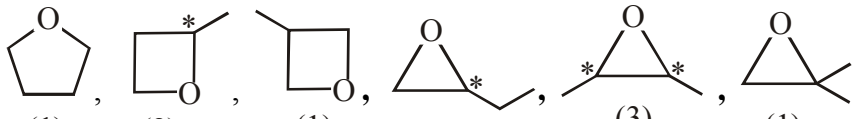
Sol. M is a organic compound known as camphor. M contains two **rigid** chiral centre so it can exist only in **two** enantiomeric forms.



15 Ans. (5) **16. Ans. (7)**

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17. Ans. (10.00)

Sol. 

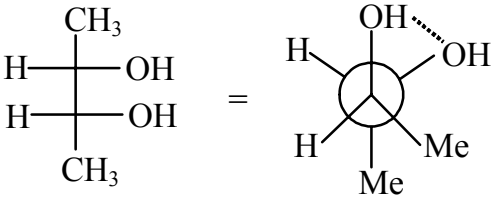
(1) (2) (1) (2) (3) (1)

(R + S) (R + S) R,R R,S S,S

18. Ans.(C)

19. Ans.(C)

20. Ans. (B)

Sol. 

Most stable conformation with H–bonding

Meso butane -2,3,-diol

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